

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412346
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Rybnikov; Silvina et al.

Methods for medical image visualization

Abstract

A computer-implemented method includes obtaining a three-dimensional (3D) image of a region of a body of a patient, the 3D image having feature values. The 3D image is segmented to define one or more regions of interest (ROIs). At least one region of interest (ROI) feature threshold is determined. A background feature threshold is determined. A 3D model is generated from the 3D image based on the determined at least one ROI feature threshold, the determined background feature threshold, and the segmentation. The 3D model is output for display to a user.

Inventors: Rybnikov; Silvina (Zichron Ya'akov, IL), Bar-Zohar; Gal (Yokneam Hamoshava, IL), Elimelech; Nissan (Be'erotaim, IL), Portal; Yotam (Kiryat Motzkin, IL)

Applicant: AUGMEDICS LTD. (Yokneam Illit, IL)

Family ID: 88419409

Assignee: AUGMEDICS LTD. (Yokneam Illit, IL)

Appl. No.: 18/399253

Filed: December 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240127559 A1	Apr. 18, 2024

Related U.S. Application Data

continuation parent-doc US 18365566 20230804 PENDING child-doc US 18399253
continuation parent-doc WO PCT/IB2023/054056 20230420 PENDING child-doc US 18365566
us-provisional-application US 63428781 20221130
us-provisional-application US 63333128 20220421

Publication Classification

Int. Cl.: G06T19/00 (20110101)

U.S. Cl.:

CPC G06T19/006 (20130101); G06T2210/41 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3101715	12/1962	Glassman	N/A	N/A
3690776	12/1971	Zaporoshan	N/A	N/A
4459358	12/1983	Berke	N/A	N/A
4711512	12/1986	Upatnieks	N/A	N/A
4863238	12/1988	Brewster	N/A	N/A
4944739	12/1989	Torre	N/A	N/A
5100420	12/1991	Green et al.	N/A	N/A
5147365	12/1991	Whitlock et al.	N/A	N/A
5357292	12/1993	Wiedner	N/A	N/A
5410802	12/1994	Buckley	N/A	N/A
5441042	12/1994	Putman	N/A	N/A
5442146	12/1994	Bell et al.	N/A	N/A
5510832	12/1995	Garcia	N/A	N/A
D370309	12/1995	Stucky	N/A	N/A
5620188	12/1996	Mccurry et al.	N/A	N/A
5636255	12/1996	Ellis	N/A	N/A
5665092	12/1996	Mangiardi et al.	N/A	N/A
5743731	12/1997	Lares et al.	N/A	N/A
5771121	12/1997	Hentschke	N/A	N/A
5792046	12/1997	Dobrovolny	N/A	N/A
5841507	12/1997	Barnes	N/A	N/A
6006126	12/1998	Cosman	N/A	N/A
6038467	12/1999	De Blik et al.	N/A	N/A
6125164	12/1999	Murphy et al.	N/A	N/A
6138530	12/1999	McClure	N/A	N/A
6147805	12/1999	Ferguson	N/A	N/A
6227667	12/2000	Halldorsson et al.	N/A	N/A
6256529	12/2000	Holupka et al.	N/A	N/A
6285505	12/2000	Melville et al.	N/A	N/A
6314310	12/2000	Ben-Haim et al.	N/A	N/A
6349001	12/2001	Spitzer	N/A	N/A
6444192	12/2001	Mattrey	N/A	N/A
6447503	12/2001	Wynne et al.	N/A	N/A

6449090	12/2001	Omar et al.	N/A	N/A
6456405	12/2001	Horikoshi et al.	N/A	N/A
6456868	12/2001	Saito et al.	N/A	N/A
6474159	12/2001	Foxlin et al.	N/A	N/A
6518939	12/2002	Kikuchi	N/A	N/A
6527777	12/2002	Justin	N/A	N/A
6529331	12/2002	Massof et al.	N/A	N/A
6549645	12/2002	Oikawa et al.	N/A	N/A
6578962	12/2002	Amir et al.	N/A	N/A
6609022	12/2002	Vilsmeier et al.	N/A	N/A
6610009	12/2002	Person	N/A	N/A
D480476	12/2002	Martinson et al.	N/A	N/A
6659611	12/2002	Amir et al.	N/A	N/A
6675040	12/2003	Cosman	N/A	N/A
6683584	12/2003	Ronzani et al.	N/A	N/A
6690964	12/2003	Bieger et al.	N/A	N/A
6714810	12/2003	Grzeszczuk et al.	N/A	N/A
6737425	12/2003	Yamamoto et al.	N/A	N/A
6740882	12/2003	Weinberg	N/A	N/A
6757068	12/2003	Foxlin	N/A	N/A
6759200	12/2003	Stanton, Jr.	N/A	N/A
6847336	12/2004	Lemelson et al.	N/A	N/A
6856324	12/2004	Sauer et al.	N/A	N/A
6856826	12/2004	Seeley et al.	N/A	N/A
6891518	12/2004	Sauer et al.	N/A	N/A
6900777	12/2004	Hebert et al.	N/A	N/A
6919867	12/2004	Sauer	N/A	N/A
6921167	12/2004	Nagata	N/A	N/A
6966668	12/2004	Cugini et al.	N/A	N/A
6980849	12/2004	Sasso	N/A	N/A
6993374	12/2005	Sasso	N/A	N/A
6997552	12/2005	Hung	N/A	N/A
6999239	12/2005	Martins et al.	N/A	N/A
7000262	12/2005	Bielefeld	N/A	N/A
7035371	12/2005	Boese et al.	N/A	N/A
7043961	12/2005	Pandey et al.	N/A	N/A
7072435	12/2005	Metz et al.	N/A	N/A
7103233	12/2005	Stearns	N/A	N/A
7107091	12/2005	Jutras et al.	N/A	N/A
7112656	12/2005	Desnoyers et al.	N/A	N/A
7141812	12/2005	Appleby et al.	N/A	N/A
7157459	12/2006	Ohta et al.	N/A	N/A
7169785	12/2006	Timmer et al.	N/A	N/A
7171255	12/2006	Holupka et al.	N/A	N/A
7176936	12/2006	Sauer et al.	N/A	N/A
7187792	12/2006	Fu et al.	N/A	N/A
7190331	12/2006	Genc et al.	N/A	N/A
7194295	12/2006	Vilsmeier	N/A	N/A
7215322	12/2006	Genc et al.	N/A	N/A
7229078	12/2006	Lehot	N/A	N/A

7231076	12/2006	Fu et al.	N/A	N/A
7235076	12/2006	Pacheco	N/A	N/A
7239330	12/2006	Sauer et al.	N/A	N/A
7241292	12/2006	Hooven	N/A	N/A
7259266	12/2006	Carter et al.	N/A	N/A
7260426	12/2006	Schweikard et al.	N/A	N/A
7269192	12/2006	Hayashi	N/A	N/A
7281826	12/2006	Huang	N/A	N/A
7315636	12/2007	Kuduvalli	N/A	N/A
7320556	12/2007	Vagn-Erik	N/A	N/A
7330578	12/2007	Wang et al.	N/A	N/A
7359535	12/2007	Salla et al.	N/A	N/A
7364314	12/2007	Nilsen et al.	N/A	N/A
7366934	12/2007	Narayan et al.	N/A	N/A
7379077	12/2007	Bani-Hashemi et al.	N/A	N/A
7431453	12/2007	Hogan	N/A	N/A
7435219	12/2007	Kim	N/A	N/A
7450743	12/2007	Sundar et al.	N/A	N/A
7458977	12/2007	McGinley et al.	N/A	N/A
7462852	12/2007	Appleby et al.	N/A	N/A
7493153	12/2008	Ahmed et al.	N/A	N/A
7505617	12/2008	Fu et al.	N/A	N/A
7507968	12/2008	Wollenweber et al.	N/A	N/A
7518136	12/2008	Appleby et al.	N/A	N/A
7525735	12/2008	Sottolare et al.	N/A	N/A
D592691	12/2008	Chang	N/A	N/A
D592692	12/2008	Chang	N/A	N/A
D592693	12/2008	Chang	N/A	N/A
7536216	12/2008	Geiger et al.	N/A	N/A
7542791	12/2008	Mire et al.	N/A	N/A
7556428	12/2008	Sukovic et al.	N/A	N/A
7557824	12/2008	Holliman	N/A	N/A
7563228	12/2008	Ma et al.	N/A	N/A
7567834	12/2008	Clayton et al.	N/A	N/A
7570791	12/2008	Frank et al.	N/A	N/A
7586686	12/2008	Hall	N/A	N/A
D602620	12/2008	Cristoforo	N/A	N/A
7605826	12/2008	Sauer	N/A	N/A
7606613	12/2008	Simon et al.	N/A	N/A
7607775	12/2008	Hermanson et al.	N/A	N/A
7620223	12/2008	Xu et al.	N/A	N/A
7623902	12/2008	Pacheco	N/A	N/A
7627085	12/2008	Boyden et al.	N/A	N/A
7630753	12/2008	Simon et al.	N/A	N/A
7633501	12/2008	Wood et al.	N/A	N/A
7645050	12/2009	Wilt et al.	N/A	N/A
7653226	12/2009	Guhring et al.	N/A	N/A
7657075	12/2009	Viswanathan	N/A	N/A
7689019	12/2009	Boese et al.	N/A	N/A
7689042	12/2009	Brunner et al.	N/A	N/A

7689320	12/2009	Prisco et al.	N/A	N/A
7699486	12/2009	Beiner	N/A	N/A
7699793	12/2009	Goette et al.	N/A	N/A
7719769	12/2009	Sugihara et al.	N/A	N/A
D617825	12/2009	Chang	N/A	N/A
7734327	12/2009	Colquhoun	N/A	N/A
D619285	12/2009	Cristoforo	N/A	N/A
7751865	12/2009	Jascob et al.	N/A	N/A
7758204	12/2009	Klipstein et al.	N/A	N/A
7768702	12/2009	Hirose et al.	N/A	N/A
7769236	12/2009	Fiala	N/A	N/A
7773074	12/2009	Arenson et al.	N/A	N/A
7774044	12/2009	Sauer et al.	N/A	N/A
7822483	12/2009	Stone et al.	N/A	N/A
D628307	12/2009	Krause-Bonte	N/A	N/A
7826902	12/2009	Stone et al.	N/A	N/A
7831073	12/2009	Fu et al.	N/A	N/A
7831096	12/2009	Williamson, Jr.	N/A	N/A
7835778	12/2009	Foley et al.	N/A	N/A
7835784	12/2009	Mire et al.	N/A	N/A
7837987	12/2009	Shi et al.	N/A	N/A
7840093	12/2009	Fu et al.	N/A	N/A
7840253	12/2009	Tremblay et al.	N/A	N/A
7840256	12/2009	Lakin et al.	N/A	N/A
7853305	12/2009	Simon et al.	N/A	N/A
7854705	12/2009	Pawluczyk et al.	N/A	N/A
7857271	12/2009	Lees	N/A	N/A
7860282	12/2009	Boese et al.	N/A	N/A
D630766	12/2010	Harbin	N/A	N/A
7865269	12/2010	Prisco et al.	N/A	N/A
7874686	12/2010	Rossner et al.	N/A	N/A
7881770	12/2010	Melkent et al.	N/A	N/A
7893413	12/2010	Appleby et al.	N/A	N/A
7894649	12/2010	Fu et al.	N/A	N/A
7920162	12/2010	Masini et al.	N/A	N/A
7922391	12/2010	Essenreiter et al.	N/A	N/A
7938553	12/2010	Beiner	N/A	N/A
7945310	12/2010	Gattani et al.	N/A	N/A
7953471	12/2010	Clayton et al.	N/A	N/A
7969383	12/2010	Eberl et al.	N/A	N/A
7974677	12/2010	Mire et al.	N/A	N/A
7985756	12/2010	Barlow et al.	N/A	N/A
7991557	12/2010	Liew et al.	N/A	N/A
7993353	12/2010	Roner et al.	N/A	N/A
7996064	12/2010	Simon et al.	N/A	N/A
8004524	12/2010	Deinzer	N/A	N/A
8021300	12/2010	Ma et al.	N/A	N/A
8022984	12/2010	Cheong et al.	N/A	N/A
8045266	12/2010	Nakamura	N/A	N/A
8060181	12/2010	Rodriguez et al.	N/A	N/A

8068581	12/2010	Boese et al.	N/A	N/A
8068896	12/2010	Daghighian et al.	N/A	N/A
8077943	12/2010	Williams et al.	N/A	N/A
8079957	12/2010	Ma et al.	N/A	N/A
8081812	12/2010	Kreiser	N/A	N/A
8085075	12/2010	Huffman et al.	N/A	N/A
8085897	12/2010	Morton	N/A	N/A
8090175	12/2011	Fu et al.	N/A	N/A
8092400	12/2011	Warkentine et al.	N/A	N/A
8108072	12/2011	Zhao et al.	N/A	N/A
8112292	12/2011	Simon	N/A	N/A
8116847	12/2011	Gattani et al.	N/A	N/A
8120847	12/2011	Chang	N/A	N/A
8121255	12/2011	Sugiyama	N/A	N/A
8155479	12/2011	Hoffman et al.	N/A	N/A
8180132	12/2011	Gorges et al.	N/A	N/A
8180429	12/2011	Sasso	N/A	N/A
8208599	12/2011	Ye et al.	N/A	N/A
8216211	12/2011	Mathis et al.	N/A	N/A
8221402	12/2011	Francischelli et al.	N/A	N/A
8239001	12/2011	Verard et al.	N/A	N/A
8244012	12/2011	Liang et al.	N/A	N/A
8253778	12/2011	Atsushi	N/A	N/A
8271069	12/2011	Jascob et al.	N/A	N/A
8280491	12/2011	Kuduvalli et al.	N/A	N/A
8285021	12/2011	Boese et al.	N/A	N/A
8300315	12/2011	Kobayashi	N/A	N/A
8305685	12/2011	Heine et al.	N/A	N/A
8306305	12/2011	Porat et al.	N/A	N/A
8309932	12/2011	Haselman et al.	N/A	N/A
8317320	12/2011	Huang	N/A	N/A
8328815	12/2011	Farr et al.	N/A	N/A
8335553	12/2011	Rubner et al.	N/A	N/A
8335557	12/2011	Maschke	N/A	N/A
8340379	12/2011	Razzaque et al.	N/A	N/A
8369925	12/2012	Giesel et al.	N/A	N/A
8386022	12/2012	Jutras et al.	N/A	N/A
8394144	12/2012	Zehavi et al.	N/A	N/A
8398541	12/2012	Dimaio et al.	N/A	N/A
8444266	12/2012	Waters	N/A	N/A
8457719	12/2012	Moctezuma De La Barrera et al.	N/A	N/A
8467851	12/2012	Mire et al.	N/A	N/A
8469902	12/2012	Dick et al.	N/A	N/A
8475470	12/2012	Von Jako	N/A	N/A
8494612	12/2012	Vetter et al.	N/A	N/A
8509503	12/2012	Nahum et al.	N/A	N/A
8511827	12/2012	Hua et al.	N/A	N/A
8531394	12/2012	Maltz	N/A	N/A
8540364	12/2012	Waters	N/A	N/A

8545012	12/2012	Waters	N/A	N/A
8548567	12/2012	Maschke et al.	N/A	N/A
8556883	12/2012	Saleh	N/A	N/A
8559596	12/2012	Thomson et al.	N/A	N/A
8567945	12/2012	Waters	N/A	N/A
8571353	12/2012	Watanabe	N/A	N/A
8585598	12/2012	Razzaque et al.	N/A	N/A
8600001	12/2012	Schweizer	N/A	N/A
8600477	12/2012	Beyar et al.	N/A	N/A
8605199	12/2012	Imai	N/A	N/A
8611988	12/2012	Miyamoto	N/A	N/A
8612024	12/2012	Stone et al.	N/A	N/A
8634897	12/2013	Simon et al.	N/A	N/A
8641621	12/2013	Razzaque et al.	N/A	N/A
8643950	12/2013	König	N/A	N/A
8644907	12/2013	Hartmann et al.	N/A	N/A
8674902	12/2013	Park et al.	N/A	N/A
8686923	12/2013	Eberl et al.	N/A	N/A
8690581	12/2013	Ruf et al.	N/A	N/A
8690776	12/2013	Razzaque et al.	N/A	N/A
8692845	12/2013	Fedorovskaya et al.	N/A	N/A
8693632	12/2013	Allison	N/A	N/A
8694075	12/2013	Groszmann et al.	N/A	N/A
8699765	12/2013	Hao et al.	N/A	N/A
8705829	12/2013	Frank et al.	N/A	N/A
8737708	12/2013	Hartmann et al.	N/A	N/A
8746887	12/2013	Shestak et al.	N/A	N/A
8764025	12/2013	Gao	N/A	N/A
8784450	12/2013	Moskowitz et al.	N/A	N/A
8786689	12/2013	Liu	N/A	N/A
D710545	12/2013	Wu	N/A	N/A
D710546	12/2013	Wu	N/A	N/A
8827934	12/2013	Chopra et al.	N/A	N/A
8831706	12/2013	Fu et al.	N/A	N/A
8836768	12/2013	Rafii et al.	N/A	N/A
8838199	12/2013	Simon et al.	N/A	N/A
8848977	12/2013	Bammer et al.	N/A	N/A
8855395	12/2013	Baturin et al.	N/A	N/A
8878900	12/2013	Yang et al.	N/A	N/A
8879815	12/2013	Miao et al.	N/A	N/A
8885177	12/2013	Ben-Yishai et al.	N/A	N/A
8890772	12/2013	Woo et al.	N/A	N/A
8890773	12/2013	Pederson	N/A	N/A
8890943	12/2013	Lee et al.	N/A	N/A
8897514	12/2013	Feikas et al.	N/A	N/A
8900131	12/2013	Chopra et al.	N/A	N/A
8903150	12/2013	Star-Lack et al.	N/A	N/A
8908952	12/2013	Isaacs et al.	N/A	N/A
8911358	12/2013	Koninckx et al.	N/A	N/A
8917268	12/2013	Johnsen et al.	N/A	N/A

8920776	12/2013	Gaiger et al.	N/A	N/A
8922589	12/2013	Laor	N/A	N/A
8941559	12/2014	Bar-Zeev et al.	N/A	N/A
8942455	12/2014	Chou et al.	N/A	N/A
8950877	12/2014	Northey et al.	N/A	N/A
8953246	12/2014	Koenig	N/A	N/A
8961500	12/2014	Dicorleto et al.	N/A	N/A
8965583	12/2014	Ortmaier et al.	N/A	N/A
8969829	12/2014	Wollenweber et al.	N/A	N/A
8989349	12/2014	Thomson et al.	N/A	N/A
8992580	12/2014	Bar et al.	N/A	N/A
8994729	12/2014	Nakamura	N/A	N/A
8994795	12/2014	Oh	N/A	N/A
9004711	12/2014	Gerolemou	N/A	N/A
9005211	12/2014	Brundobler et al.	N/A	N/A
9011441	12/2014	Bertagnoli et al.	N/A	N/A
9057759	12/2014	Klingenbeck et al.	N/A	N/A
9060757	12/2014	Lawson et al.	N/A	N/A
9066751	12/2014	Sasso	N/A	N/A
9081436	12/2014	Berme et al.	N/A	N/A
9084635	12/2014	Nuckley et al.	N/A	N/A
9085643	12/2014	Svanborg et al.	N/A	N/A
9087471	12/2014	Miao	N/A	N/A
9100643	12/2014	McDowall et al.	N/A	N/A
9101394	12/2014	Arata et al.	N/A	N/A
9104902	12/2014	Xu et al.	N/A	N/A
9111175	12/2014	Strommer et al.	N/A	N/A
9123155	12/2014	Cunningham et al.	N/A	N/A
9125556	12/2014	Zehavi et al.	N/A	N/A
9129054	12/2014	Nawana et al.	N/A	N/A
9129372	12/2014	Kriston et al.	N/A	N/A
9132361	12/2014	Smithwick	N/A	N/A
9135706	12/2014	Zagorchev et al.	N/A	N/A
9141873	12/2014	Takemoto	N/A	N/A
9142020	12/2014	Deguisse et al.	N/A	N/A
9149317	12/2014	Arthur et al.	N/A	N/A
9165203	12/2014	McCarthy	N/A	N/A
9165362	12/2014	Siewerdsen et al.	N/A	N/A
9179984	12/2014	Teichman et al.	N/A	N/A
D746354	12/2014	Chang	N/A	N/A
9208916	12/2014	Appleby et al.	N/A	N/A
9220573	12/2014	Kendrick et al.	N/A	N/A
9225895	12/2014	Kozinski	N/A	N/A
9232982	12/2015	Soler et al.	N/A	N/A
9235934	12/2015	Mandella et al.	N/A	N/A
9240046	12/2015	Carrell et al.	N/A	N/A
9244278	12/2015	Sugiyama et al.	N/A	N/A
9247240	12/2015	Park et al.	N/A	N/A
9259192	12/2015	Ishihara	N/A	N/A
9265572	12/2015	Fuchs et al.	N/A	N/A

9269192	12/2015	Kobayashi	N/A	N/A
9283052	12/2015	Rodriguez Ponce	N/A	N/A
9286730	12/2015	Bar-Zeev et al.	N/A	N/A
9289267	12/2015	Sauer et al.	N/A	N/A
9294222	12/2015	Proctor, Jr.	N/A	N/A
9300949	12/2015	Ahearn	N/A	N/A
9305354	12/2015	Burlon et al.	N/A	N/A
9310591	12/2015	Hua et al.	N/A	N/A
9320474	12/2015	Demri et al.	N/A	N/A
9323055	12/2015	Baillot	N/A	N/A
9330477	12/2015	Rappel	N/A	N/A
9335547	12/2015	Takano et al.	N/A	N/A
9335567	12/2015	Nakamura	N/A	N/A
9341704	12/2015	Picard et al.	N/A	N/A
9344686	12/2015	Moharir	N/A	N/A
9349066	12/2015	Koo et al.	N/A	N/A
9349520	12/2015	Demetriou et al.	N/A	N/A
9364294	12/2015	Razzaque et al.	N/A	N/A
9370332	12/2015	Paladini et al.	N/A	N/A
9373166	12/2015	Azar	N/A	N/A
9375639	12/2015	Kobayashi et al.	N/A	N/A
9378558	12/2015	Kajiwara et al.	N/A	N/A
9380287	12/2015	Nistico et al.	N/A	N/A
9387008	12/2015	Sarvestani et al.	N/A	N/A
9392129	12/2015	Simmons	N/A	N/A
9395542	12/2015	Tilleman et al.	N/A	N/A
9398936	12/2015	Razzaque et al.	N/A	N/A
9400384	12/2015	Griffith	N/A	N/A
9414041	12/2015	Ko et al.	N/A	N/A
9424611	12/2015	Kanjirathinkal et al.	N/A	N/A
9424641	12/2015	Wiemker et al.	N/A	N/A
9427286	12/2015	Siewerdsen et al.	N/A	N/A
9438894	12/2015	Park et al.	N/A	N/A
9443488	12/2015	Borenstein et al.	N/A	N/A
9453804	12/2015	Tahtali	N/A	N/A
9456878	12/2015	MacFarlane et al.	N/A	N/A
9465235	12/2015	Chang	N/A	N/A
9468373	12/2015	Larsen	N/A	N/A
9470908	12/2015	Frankel et al.	N/A	N/A
9473766	12/2015	Douglas et al.	N/A	N/A
9492222	12/2015	Singh	N/A	N/A
9495585	12/2015	Bicer et al.	N/A	N/A
9498132	12/2015	Maier-Hein et al.	N/A	N/A
9498231	12/2015	Haider et al.	N/A	N/A
9499999	12/2015	Nanqing	N/A	N/A
9507155	12/2015	Morimoto	N/A	N/A
9513495	12/2015	Waters	N/A	N/A
9521966	12/2015	Schwartz	N/A	N/A
9526443	12/2015	Berme et al.	N/A	N/A
9530382	12/2015	Simmons	N/A	N/A

9532846	12/2016	Nakamura	N/A	N/A
9532849	12/2016	Anderson et al.	N/A	N/A
9533407	12/2016	Ragner	N/A	N/A
9538962	12/2016	Hannaford et al.	N/A	N/A
9545233	12/2016	Sirpad et al.	N/A	N/A
9546779	12/2016	Rementer	N/A	N/A
9547174	12/2016	Gao et al.	N/A	N/A
9547940	12/2016	Sun et al.	N/A	N/A
9557566	12/2016	Fujimaki	N/A	N/A
9560318	12/2016	Reina et al.	N/A	N/A
9561095	12/2016	Nguyen et al.	N/A	N/A
9561446	12/2016	Brecher	N/A	N/A
9565415	12/2016	Zhang et al.	N/A	N/A
9572661	12/2016	Robin et al.	N/A	N/A
9576398	12/2016	Zehner et al.	N/A	N/A
9576556	12/2016	Simmons	N/A	N/A
9581822	12/2016	Morimoto	N/A	N/A
9610056	12/2016	Lavallee et al.	N/A	N/A
9612657	12/2016	Bertram et al.	N/A	N/A
9626936	12/2016	Bell	N/A	N/A
9629595	12/2016	Walker et al.	N/A	N/A
9633431	12/2016	Merlet	N/A	N/A
9645395	12/2016	Bolas et al.	N/A	N/A
9646423	12/2016	Sun et al.	N/A	N/A
9672597	12/2016	Amiot et al.	N/A	N/A
9672607	12/2016	Demri et al.	N/A	N/A
9672640	12/2016	Kleiner	N/A	N/A
9675306	12/2016	Morton	N/A	N/A
9675319	12/2016	Razzaque et al.	N/A	N/A
9684980	12/2016	Royalty et al.	N/A	N/A
9690119	12/2016	Garofolo et al.	N/A	N/A
RE46463	12/2016	Fienbloom et al.	N/A	N/A
9693748	12/2016	Rai et al.	N/A	N/A
9710968	12/2016	Dillavou et al.	N/A	N/A
9713502	12/2016	Finkman et al.	N/A	N/A
9724119	12/2016	Hissong et al.	N/A	N/A
9724165	12/2016	Arata et al.	N/A	N/A
9726888	12/2016	Giartosio et al.	N/A	N/A
9728006	12/2016	Varga	N/A	N/A
9729831	12/2016	Birnkrant et al.	N/A	N/A
9746739	12/2016	Alton et al.	N/A	N/A
9757034	12/2016	Desjardins et al.	N/A	N/A
9757087	12/2016	Simon et al.	N/A	N/A
9766441	12/2016	Rappel	N/A	N/A
9766459	12/2016	Alton et al.	N/A	N/A
9767608	12/2016	Lee et al.	N/A	N/A
9770203	12/2016	Berme et al.	N/A	N/A
9772102	12/2016	Ferguson	N/A	N/A
9772495	12/2016	Tam et al.	N/A	N/A
9791138	12/2016	Feinbloom et al.	N/A	N/A

9800995	12/2016	Libin et al.	N/A	N/A
9805504	12/2016	Zhang et al.	N/A	N/A
9808148	12/2016	Miller et al.	N/A	N/A
9839448	12/2016	Reckling et al.	N/A	N/A
9844413	12/2016	Daon et al.	N/A	N/A
9851080	12/2016	Wilt et al.	N/A	N/A
9858663	12/2017	Penney et al.	N/A	N/A
9861446	12/2017	Lang	N/A	N/A
9864214	12/2017	Fass	N/A	N/A
9872733	12/2017	Shoham et al.	N/A	N/A
9875544	12/2017	Rai et al.	N/A	N/A
9877642	12/2017	Duret	N/A	N/A
9885465	12/2017	Nguyen	N/A	N/A
9886552	12/2017	Dillavou et al.	N/A	N/A
9886760	12/2017	Liu et al.	N/A	N/A
9892564	12/2017	Cvetko et al.	N/A	N/A
9898866	12/2017	Fuchs et al.	N/A	N/A
9901414	12/2017	Lively et al.	N/A	N/A
9911187	12/2017	Steinle et al.	N/A	N/A
9911236	12/2017	Bar et al.	N/A	N/A
9927611	12/2017	Rudy et al.	N/A	N/A
9928629	12/2017	Benishti et al.	N/A	N/A
9940750	12/2017	Dillavou et al.	N/A	N/A
9943374	12/2017	Merritt et al.	N/A	N/A
9947110	12/2017	Haimerl	N/A	N/A
9952664	12/2017	Border et al.	N/A	N/A
9956054	12/2017	Aguirre-Valencia	N/A	N/A
9958674	12/2017	Border	N/A	N/A
9959620	12/2017	Merlet	N/A	N/A
9959629	12/2017	Dillavou et al.	N/A	N/A
9965681	12/2017	Border et al.	N/A	N/A
9968297	12/2017	Connor	N/A	N/A
9980780	12/2017	Lang	N/A	N/A
9986228	12/2017	Woods	N/A	N/A
D824523	12/2017	Paoli et al.	N/A	N/A
10010379	12/2017	Gibby et al.	N/A	N/A
10013531	12/2017	Richards et al.	N/A	N/A
10015243	12/2017	Kazerani et al.	N/A	N/A
10016243	12/2017	Esterberg	N/A	N/A
10022064	12/2017	Kim et al.	N/A	N/A
10022065	12/2017	Ben-Yishai et al.	N/A	N/A
10022104	12/2017	Sell et al.	N/A	N/A
10023615	12/2017	Bonny	N/A	N/A
10026015	12/2017	Cavusoglu et al.	N/A	N/A
10034713	12/2017	Yang et al.	N/A	N/A
10042167	12/2017	Mcdowall et al.	N/A	N/A
10046165	12/2017	Frewin et al.	N/A	N/A
10055838	12/2017	Elenbaas et al.	N/A	N/A
10066816	12/2017	Chang	N/A	N/A
10067359	12/2017	Ushakov	N/A	N/A

10073515	12/2017	Awdeh	N/A	N/A
10080616	12/2017	Wilkinson et al.	N/A	N/A
10082680	12/2017	Chung	N/A	N/A
10085709	12/2017	Lavallee et al.	N/A	N/A
10105187	12/2017	Corndorf et al.	N/A	N/A
10107483	12/2017	Oren	N/A	N/A
10108833	12/2017	Hong et al.	N/A	N/A
10123840	12/2017	Dorman	N/A	N/A
10130378	12/2017	Bryan	N/A	N/A
10132483	12/2017	Feinbloom et al.	N/A	N/A
10134166	12/2017	Benishti et al.	N/A	N/A
10134194	12/2017	Kepner et al.	N/A	N/A
10139652	12/2017	Windham	N/A	N/A
10139920	12/2017	Isaacs et al.	N/A	N/A
10142496	12/2017	Rao et al.	N/A	N/A
10151928	12/2017	Ushakov	N/A	N/A
10154239	12/2017	Casas	N/A	N/A
10159530	12/2017	Lang	N/A	N/A
10163207	12/2017	Merlet	N/A	N/A
10166079	12/2018	McLachlin et al.	N/A	N/A
10175507	12/2018	Nakamura	N/A	N/A
10175753	12/2018	Boesen	N/A	N/A
10181361	12/2018	Dillavou et al.	N/A	N/A
10186055	12/2018	Takahashi et al.	N/A	N/A
10188672	12/2018	Wagner	N/A	N/A
10194131	12/2018	Casas	N/A	N/A
10194990	12/2018	Amanatullah et al.	N/A	N/A
10194993	12/2018	Roger et al.	N/A	N/A
10195076	12/2018	Fateh	N/A	N/A
10197803	12/2018	Badiali et al.	N/A	N/A
10197816	12/2018	Waisman et al.	N/A	N/A
10207315	12/2018	Appleby et al.	N/A	N/A
10212517	12/2018	Beltran et al.	N/A	N/A
10230719	12/2018	Vaughn et al.	N/A	N/A
10231893	12/2018	Lei et al.	N/A	N/A
10235606	12/2018	Miao et al.	N/A	N/A
10240769	12/2018	Braganca et al.	N/A	N/A
10247965	12/2018	Ton	N/A	N/A
10251724	12/2018	McLachlin et al.	N/A	N/A
10261324	12/2018	Chuang et al.	N/A	N/A
10262424	12/2018	Ketcha et al.	N/A	N/A
10274731	12/2018	Maimone	N/A	N/A
10278777	12/2018	Lang	N/A	N/A
10292768	12/2018	Lang	N/A	N/A
10296805	12/2018	Yang et al.	N/A	N/A
10319154	12/2018	Chakravarthula et al.	N/A	N/A
10326975	12/2018	Casas	N/A	N/A
10332267	12/2018	Rai et al.	N/A	N/A
10339719	12/2018	Jagga et al.	N/A	N/A
10352543	12/2018	Braganca et al.	N/A	N/A

10357146	12/2018	Fiebel et al.	N/A	N/A
10357574	12/2018	Hilderbrand et al.	N/A	N/A
10366489	12/2018	Boettger et al.	N/A	N/A
10368947	12/2018	Lang	N/A	N/A
10368948	12/2018	Tripathi	N/A	N/A
10382748	12/2018	Benishti et al.	N/A	N/A
10383654	12/2018	Yilmaz et al.	N/A	N/A
10386645	12/2018	Abou Shousha	N/A	N/A
10388076	12/2018	Bar-Zeev et al.	N/A	N/A
10398514	12/2018	Ryan et al.	N/A	N/A
10401657	12/2018	Jiang et al.	N/A	N/A
10405825	12/2018	Rai et al.	N/A	N/A
10405927	12/2018	Lang	N/A	N/A
10413752	12/2018	Berlinger et al.	N/A	N/A
10419655	12/2018	Sivan	N/A	N/A
10420626	12/2018	Tokuda et al.	N/A	N/A
10420813	12/2018	Newell-Rogers et al.	N/A	N/A
10424115	12/2018	Ellerbrock	N/A	N/A
D862469	12/2018	Sadot et al.	N/A	N/A
10426554	12/2018	Siewerdsen et al.	N/A	N/A
10429675	12/2018	Greget	N/A	N/A
10431008	12/2018	Djajadiningrat et al.	N/A	N/A
10433814	12/2018	Razzaque et al.	N/A	N/A
10434335	12/2018	Takahashi et al.	N/A	N/A
10441236	12/2018	Bar-Tal et al.	N/A	N/A
10444514	12/2018	Abou Shousha et al.	N/A	N/A
10447947	12/2018	Liu	N/A	N/A
10448003	12/2018	Grafenberg	N/A	N/A
10449040	12/2018	Lashinski et al.	N/A	N/A
10453187	12/2018	Peterson et al.	N/A	N/A
10463434	12/2018	Siegler et al.	N/A	N/A
10465892	12/2018	Feinbloom et al.	N/A	N/A
10466487	12/2018	Blum et al.	N/A	N/A
10470732	12/2018	Baumgart et al.	N/A	N/A
10473314	12/2018	Braganca et al.	N/A	N/A
10485989	12/2018	Jordan et al.	N/A	N/A
10488663	12/2018	Choi	N/A	N/A
D869772	12/2018	Gand	N/A	N/A
D870977	12/2018	Berggren et al.	N/A	N/A
10492755	12/2018	Lin et al.	N/A	N/A
10499997	12/2018	Weinstein et al.	N/A	N/A
10502363	12/2018	Edwards et al.	N/A	N/A
10504231	12/2018	Fiala	N/A	N/A
10507066	12/2018	Dimaio et al.	N/A	N/A
10511822	12/2018	Casas	N/A	N/A
10517544	12/2018	Taguchi et al.	N/A	N/A
10537395	12/2019	Perez	N/A	N/A
10540780	12/2019	Cousins et al.	N/A	N/A
10543485	12/2019	Ismagilov et al.	N/A	N/A
10546423	12/2019	Jones et al.	N/A	N/A

10548557	12/2019	Lim et al.	N/A	N/A
10555775	12/2019	Hoffman et al.	N/A	N/A
10568535	12/2019	Roberts et al.	N/A	N/A
10571696	12/2019	Urey et al.	N/A	N/A
10571716	12/2019	Chapiro	N/A	N/A
10573086	12/2019	Bar-Zeev et al.	N/A	N/A
10573087	12/2019	Gallop et al.	N/A	N/A
10577630	12/2019	Zhang et al.	N/A	N/A
10586400	12/2019	Douglas	N/A	N/A
10591737	12/2019	Yildiz et al.	N/A	N/A
10592748	12/2019	Cousins et al.	N/A	N/A
10594998	12/2019	Casas	N/A	N/A
10595716	12/2019	Nazareth et al.	N/A	N/A
10601950	12/2019	Devam et al.	N/A	N/A
10602114	12/2019	Casas	N/A	N/A
10603113	12/2019	Lang	N/A	N/A
10603133	12/2019	Wang et al.	N/A	N/A
10606085	12/2019	Toyama	N/A	N/A
10610172	12/2019	Hummel et al.	N/A	N/A
10610179	12/2019	Altmann	N/A	N/A
10613352	12/2019	Knoll	N/A	N/A
10617566	12/2019	Esmonde	N/A	N/A
10620460	12/2019	Carabin	N/A	N/A
10621738	12/2019	Miao et al.	N/A	N/A
10625099	12/2019	Takahashi et al.	N/A	N/A
10626473	12/2019	Mariani et al.	N/A	N/A
10631905	12/2019	Asfora et al.	N/A	N/A
10631907	12/2019	Zucker et al.	N/A	N/A
10634331	12/2019	Feinbloom et al.	N/A	N/A
10634921	12/2019	Blum et al.	N/A	N/A
10638080	12/2019	Ovchinnikov et al.	N/A	N/A
10646285	12/2019	Siemionow et al.	N/A	N/A
10650513	12/2019	Penney et al.	N/A	N/A
10650594	12/2019	Jones et al.	N/A	N/A
10652525	12/2019	Woods	N/A	N/A
10653495	12/2019	Gregerson et al.	N/A	N/A
10660715	12/2019	Dozeman	N/A	N/A
10663738	12/2019	Carlvik et al.	N/A	N/A
10665033	12/2019	Bar-Zeev et al.	N/A	N/A
10670937	12/2019	Alton et al.	N/A	N/A
10672145	12/2019	Albiol et al.	N/A	N/A
10682112	12/2019	Pizaine et al.	N/A	N/A
10682767	12/2019	Grafenberg et al.	N/A	N/A
10687901	12/2019	Thomas	N/A	N/A
10691397	12/2019	Clements	N/A	N/A
10702713	12/2019	Mori et al.	N/A	N/A
10706540	12/2019	Merlet	N/A	N/A
10709398	12/2019	Schweizer	N/A	N/A
10713801	12/2019	Jordan et al.	N/A	N/A
10716643	12/2019	Justin et al.	N/A	N/A

10722733	12/2019	Takahashi	N/A	N/A
10725535	12/2019	Yu	N/A	N/A
10731832	12/2019	Koo	N/A	N/A
10732721	12/2019	Clements	N/A	N/A
10742949	12/2019	Casas	N/A	N/A
10743939	12/2019	Lang	N/A	N/A
10743943	12/2019	Razeto et al.	N/A	N/A
10747315	12/2019	Tungare et al.	N/A	N/A
10748319	12/2019	Tao et al.	N/A	N/A
10758315	12/2019	Johnson et al.	N/A	N/A
10777094	12/2019	Rao et al.	N/A	N/A
10777315	12/2019	Zehavi et al.	N/A	N/A
10781482	12/2019	Gubatayao et al.	N/A	N/A
10792110	12/2019	Leung et al.	N/A	N/A
10799145	12/2019	West et al.	N/A	N/A
10799296	12/2019	Lang	N/A	N/A
10799298	12/2019	Crawford et al.	N/A	N/A
10799316	12/2019	Sela et al.	N/A	N/A
10810799	12/2019	Tepper et al.	N/A	N/A
10818019	12/2019	Piat et al.	N/A	N/A
10818101	12/2019	Gallop et al.	N/A	N/A
10818199	12/2019	Buras et al.	N/A	N/A
10825563	12/2019	Gibby et al.	N/A	N/A
10827164	12/2019	Perreault et al.	N/A	N/A
10831943	12/2019	Santarone et al.	N/A	N/A
10835296	12/2019	Elimelech et al.	N/A	N/A
10838206	12/2019	Fortin-Deschnes et al.	N/A	N/A
10839629	12/2019	Jones et al.	N/A	N/A
10839956	12/2019	Beydoun et al.	N/A	N/A
10841556	12/2019	Casas	N/A	N/A
10842002	12/2019	Chang	N/A	N/A
10842461	12/2019	Johnson et al.	N/A	N/A
10849691	12/2019	Zucker et al.	N/A	N/A
10849693	12/2019	Lang	N/A	N/A
10849710	12/2019	Liu	N/A	N/A
10861236	12/2019	Geri et al.	N/A	N/A
10865220	12/2019	Ebetino et al.	N/A	N/A
10869517	12/2019	Halpern	N/A	N/A
10869727	12/2019	Yanof et al.	N/A	N/A
10872472	12/2019	Watola et al.	N/A	N/A
10877262	12/2019	Luxembourg	N/A	N/A
10877296	12/2019	Lindsey et al.	N/A	N/A
10878639	12/2019	Douglas et al.	N/A	N/A
10893260	12/2020	Trail et al.	N/A	N/A
10895742	12/2020	Schneider et al.	N/A	N/A
10895743	12/2020	Dausmann	N/A	N/A
10895906	12/2020	West et al.	N/A	N/A
10898151	12/2020	Harding et al.	N/A	N/A
10908420	12/2020	Lee et al.	N/A	N/A
10921595	12/2020	Rakshit et al.	N/A	N/A

10921613	12/2020	Gupta et al.	N/A	N/A
10928321	12/2020	Rawle	N/A	N/A
10928638	12/2020	Ninan et al.	N/A	N/A
10929670	12/2020	Troy et al.	N/A	N/A
10935815	12/2020	Castaeda	N/A	N/A
10935816	12/2020	Ban et al.	N/A	N/A
10936537	12/2020	Huston	N/A	N/A
10939973	12/2020	Dimaio et al.	N/A	N/A
10939977	12/2020	Messinger et al.	N/A	N/A
10941933	12/2020	Ferguson	N/A	N/A
10946108	12/2020	Zhang et al.	N/A	N/A
10950338	12/2020	Douglas	N/A	N/A
10951872	12/2020	Casas	N/A	N/A
10964095	12/2020	Douglas	N/A	N/A
10964124	12/2020	Douglas	N/A	N/A
10966768	12/2020	Poulos	N/A	N/A
10969587	12/2020	Mcdowall et al.	N/A	N/A
10993754	12/2020	Kuntz et al.	N/A	N/A
11000335	12/2020	Dorman	N/A	N/A
11002994	12/2020	Jiang et al.	N/A	N/A
11006093	12/2020	Hegyi	N/A	N/A
11013550	12/2020	Rioux et al.	N/A	N/A
11013560	12/2020	Lang	N/A	N/A
11013562	12/2020	Marti et al.	N/A	N/A
11013573	12/2020	Chang	N/A	N/A
11013900	12/2020	Malek et al.	N/A	N/A
11016302	12/2020	Freeman et al.	N/A	N/A
11019988	12/2020	Fiebel et al.	N/A	N/A
11027027	12/2020	Manning et al.	N/A	N/A
11029147	12/2020	Abovitz et al.	N/A	N/A
11030809	12/2020	Wang	N/A	N/A
11041173	12/2020	Zhang et al.	N/A	N/A
11045663	12/2020	Mori et al.	N/A	N/A
11049293	12/2020	Chae et al.	N/A	N/A
11049476	12/2020	Fuchs et al.	N/A	N/A
11050990	12/2020	Casas	N/A	N/A
11057505	12/2020	Dharmatilleke	N/A	N/A
11058390	12/2020	Douglas	N/A	N/A
11061257	12/2020	Hakim	N/A	N/A
11064904	12/2020	Kay et al.	N/A	N/A
11065062	12/2020	Frushour et al.	N/A	N/A
11067387	12/2020	Marell et al.	N/A	N/A
11071497	12/2020	Hallack et al.	N/A	N/A
11079596	12/2020	Hua et al.	N/A	N/A
11087039	12/2020	Duff et al.	N/A	N/A
11090019	12/2020	Siemionow et al.	N/A	N/A
11097129	12/2020	Sakata et al.	N/A	N/A
11099376	12/2020	Steier et al.	N/A	N/A
11103320	12/2020	Leboeuf et al.	N/A	N/A
D930162	12/2020	Cremer et al.	N/A	N/A

11109762	12/2020	Steier et al.	N/A	N/A
11112611	12/2020	Kessler et al.	N/A	N/A
11122164	12/2020	Gigante	N/A	N/A
11123604	12/2020	Fung	N/A	N/A
11129562	12/2020	Roberts et al.	N/A	N/A
11132055	12/2020	Jones et al.	N/A	N/A
11135015	12/2020	Crawford et al.	N/A	N/A
11135016	12/2020	Frielinghaus et al.	N/A	N/A
11137610	12/2020	Kessler et al.	N/A	N/A
11141221	12/2020	Hobeika et al.	N/A	N/A
11153549	12/2020	Casas	N/A	N/A
11153555	12/2020	Healy et al.	N/A	N/A
11163176	12/2020	Karafin et al.	N/A	N/A
11164324	12/2020	Liu et al.	N/A	N/A
11166006	12/2020	Hegyi	N/A	N/A
11169380	12/2020	Manly et al.	N/A	N/A
11172990	12/2020	Lang	N/A	N/A
11179136	12/2020	Kohli et al.	N/A	N/A
11180557	12/2020	Noelle	N/A	N/A
11181747	12/2020	Kessler et al.	N/A	N/A
11185891	12/2020	Cousins et al.	N/A	N/A
11187907	12/2020	Osterman et al.	N/A	N/A
11202682	12/2020	Staunton et al.	N/A	N/A
11207150	12/2020	Healy et al.	N/A	N/A
11217028	12/2021	Jones et al.	N/A	N/A
11224483	12/2021	Steinberg et al.	N/A	N/A
11224763	12/2021	Takahashi et al.	N/A	N/A
11227417	12/2021	Berlinger et al.	N/A	N/A
11231787	12/2021	Isaacs et al.	N/A	N/A
11243404	12/2021	Mcdowall et al.	N/A	N/A
11244508	12/2021	Kazanzides et al.	N/A	N/A
11253216	12/2021	Crawford et al.	N/A	N/A
11253323	12/2021	Hughes et al.	N/A	N/A
11257190	12/2021	Mao et al.	N/A	N/A
11257241	12/2021	Tao	N/A	N/A
11263772	12/2021	Siemionow et al.	N/A	N/A
11269401	12/2021	West et al.	N/A	N/A
11272151	12/2021	Casas	N/A	N/A
11278359	12/2021	Siemionow et al.	N/A	N/A
11278413	12/2021	Lang	N/A	N/A
11280480	12/2021	Wilt et al.	N/A	N/A
11284846	12/2021	Graumann et al.	N/A	N/A
11291521	12/2021	Im	N/A	N/A
11294167	12/2021	Ishimoda	N/A	N/A
11297285	12/2021	Pierce	N/A	N/A
11300252	12/2021	Nguyen	N/A	N/A
11300790	12/2021	Cheng et al.	N/A	N/A
11304621	12/2021	Merschon et al.	N/A	N/A
11304759	12/2021	Kovtun et al.	N/A	N/A
11307402	12/2021	Steier et al.	N/A	N/A

11308663	12/2021	Alhrishy et al.	N/A	N/A
11311341	12/2021	Lang	N/A	N/A
11317973	12/2021	Calloway et al.	N/A	N/A
11337763	12/2021	Choi	N/A	N/A
11348257	12/2021	Lang	N/A	N/A
11350072	12/2021	Quiles Casas	N/A	N/A
11350965	12/2021	Yilmaz et al.	N/A	N/A
11351006	12/2021	Aferzon et al.	N/A	N/A
11354813	12/2021	Piat et al.	N/A	N/A
11360315	12/2021	Tu et al.	N/A	N/A
11373342	12/2021	Stafford et al.	N/A	N/A
11382699	12/2021	Wassall et al.	N/A	N/A
11382700	12/2021	Calloway et al.	N/A	N/A
11382712	12/2021	Elimelech et al.	N/A	N/A
11382713	12/2021	Healy et al.	N/A	N/A
11389252	12/2021	Gera et al.	N/A	N/A
11393229	12/2021	Zhou et al.	N/A	N/A
11399895	12/2021	Soper et al.	N/A	N/A
11402524	12/2021	Song et al.	N/A	N/A
11406338	12/2021	Tolkowsky	N/A	N/A
11412202	12/2021	Hegyi	N/A	N/A
11423554	12/2021	Borsdorf et al.	N/A	N/A
11430203	12/2021	Navab et al.	N/A	N/A
11432828	12/2021	Lang	N/A	N/A
11432931	12/2021	Lang	N/A	N/A
11443428	12/2021	Petersen et al.	N/A	N/A
11443431	12/2021	Flossmann et al.	N/A	N/A
11452568	12/2021	Lang	N/A	N/A
11452570	12/2021	Tolkowsky	N/A	N/A
11460915	12/2021	Frielinghaus et al.	N/A	N/A
11461936	12/2021	Freeman et al.	N/A	N/A
11461983	12/2021	Jones et al.	N/A	N/A
11464580	12/2021	Kemp et al.	N/A	N/A
11464581	12/2021	Calloway	N/A	N/A
11475625	12/2021	Douglas	N/A	N/A
11478214	12/2021	Siewerdsen et al.	N/A	N/A
11483532	12/2021	Quiles Casas	N/A	N/A
11488021	12/2021	Sun et al.	N/A	N/A
11490986	12/2021	BEEn-Yishai	N/A	N/A
11510750	12/2021	Dulin et al.	N/A	N/A
11513358	12/2021	Mcdowall et al.	N/A	N/A
11527002	12/2021	Govari	N/A	N/A
11528393	12/2021	Garofolo et al.	N/A	N/A
11544031	12/2022	Harviainen	N/A	N/A
11573420	12/2022	Sarma et al.	N/A	N/A
11589927	12/2022	Oezbek et al.	N/A	N/A
11627924	12/2022	Alexandroni et al.	N/A	N/A
11644675	12/2022	Manly et al.	N/A	N/A
11648016	12/2022	Hathaway et al.	N/A	N/A
11651499	12/2022	Wang et al.	N/A	N/A

11657518	12/2022	Ketcha et al.	N/A	N/A
11666458	12/2022	Kim et al.	N/A	N/A
11669984	12/2022	Siewerdsen et al.	N/A	N/A
11686947	12/2022	Loyola et al.	N/A	N/A
11699236	12/2022	Avital et al.	N/A	N/A
11712582	12/2022	Miyazaki et al.	N/A	N/A
11715210	12/2022	Haslam et al.	N/A	N/A
11719941	12/2022	Russell	N/A	N/A
11730389	12/2022	Farshad et al.	N/A	N/A
11733516	12/2022	Edwin et al.	N/A	N/A
11734901	12/2022	Jones et al.	N/A	N/A
11744657	12/2022	Leboeuf et al.	N/A	N/A
11750794	12/2022	Benishti et al.	N/A	N/A
11766296	12/2022	Wolf et al.	N/A	N/A
11798178	12/2022	Merlet	N/A	N/A
11801097	12/2022	Crawford et al.	N/A	N/A
11801115	12/2022	Elimelech et al.	N/A	N/A
11808943	12/2022	Robaina et al.	N/A	N/A
11815683	12/2022	Sears et al.	N/A	N/A
11826111	12/2022	Mahfouz	N/A	N/A
11832886	12/2022	Dorman	N/A	N/A
11838493	12/2022	Healy et al.	N/A	N/A
11839433	12/2022	Schaewe et al.	N/A	N/A
11839501	12/2022	Takahashi et al.	N/A	N/A
11864934	12/2023	Junio et al.	N/A	N/A
11885752	12/2023	St-Aubin et al.	N/A	N/A
11892647	12/2023	Hung et al.	N/A	N/A
11896445	12/2023	Gera et al.	N/A	N/A
11900620	12/2023	Lalys et al.	N/A	N/A
11914155	12/2023	Zhu et al.	N/A	N/A
11918310	12/2023	Roh et al.	N/A	N/A
11922631	12/2023	Haslam et al.	N/A	N/A
11941814	12/2023	Crawford et al.	N/A	N/A
11944508	12/2023	Cowin et al.	N/A	N/A
11948265	12/2023	Gibby et al.	N/A	N/A
11950968	12/2023	Wiggermann	N/A	N/A
11957420	12/2023	Lang	N/A	N/A
11961193	12/2023	Pelzl et al.	N/A	N/A
11963723	12/2023	Mlsmeier et al.	N/A	N/A
11972582	12/2023	Yan et al.	N/A	N/A
11974819	12/2023	Finley et al.	N/A	N/A
11974887	12/2023	Elimelech et al.	N/A	N/A
11977232	12/2023	Wu et al.	N/A	N/A
11980429	12/2023	Wolf et al.	N/A	N/A
11980506	12/2023	Wolf et al.	N/A	N/A
11980507	12/2023	Elimelech et al.	N/A	N/A
11980508	12/2023	Elimelech et al.	N/A	N/A
11983824	12/2023	Avisar et al.	N/A	N/A
12002171	12/2023	Jones et al.	N/A	N/A
12010285	12/2023	Quiles Casas	N/A	N/A

12014497	12/2023	Hong et al.	N/A	N/A
12019314	12/2023	Steines et al.	N/A	N/A
12026897	12/2023	Frantz et al.	N/A	N/A
12033322	12/2023	Laaksonen et al.	N/A	N/A
12044856	12/2023	Gera et al.	N/A	N/A
12044858	12/2023	Gera et al.	N/A	N/A
12053247	12/2023	Chiou	N/A	N/A
12056830	12/2023	Cvetko et al.	N/A	N/A
12059281	12/2023	Weingarten et al.	N/A	N/A
12063338	12/2023	Quiles Casas	N/A	N/A
12063345	12/2023	Benishti et al.	N/A	N/A
12069233	12/2023	Benishti et al.	N/A	N/A
12076158	12/2023	Geiger et al.	N/A	N/A
12076196	12/2023	Elimelech et al.	N/A	N/A
12079385	12/2023	Ben-Yishai et al.	N/A	N/A
12112483	12/2023	Grady et al.	N/A	N/A
12114933	12/2023	Seo et al.	N/A	N/A
12115028	12/2023	Dulin et al.	N/A	N/A
12127800	12/2023	Qian et al.	N/A	N/A
12133772	12/2023	Calloway et al.	N/A	N/A
12136176	12/2023	Spaas et al.	N/A	N/A
12142365	12/2023	Kaethner et al.	N/A	N/A
12150821	12/2023	Gera et al.	N/A	N/A
12178666	12/2023	Wolf et al.	N/A	N/A
12186028	12/2024	Gera et al.	N/A	N/A
12201384	12/2024	Wolf et al.	N/A	N/A
12206837	12/2024	Benishti et al.	N/A	N/A
12239385	12/2024	Wolf et al.	N/A	N/A
2002/0082498	12/2001	Wendt et al.	N/A	N/A
2003/0059097	12/2002	Abovitz et al.	N/A	N/A
2003/0117393	12/2002	Sauer et al.	N/A	N/A
2003/0130576	12/2002	Seeley et al.	N/A	N/A
2003/0156144	12/2002	Morita	N/A	N/A
2003/0210812	12/2002	Khamene et al.	N/A	N/A
2003/0225329	12/2002	Rossner et al.	N/A	N/A
2004/0019263	12/2003	Jutras et al.	N/A	N/A
2004/0030237	12/2003	Lee et al.	N/A	N/A
2004/0138556	12/2003	Cosman	N/A	N/A
2004/0152955	12/2003	McGinley et al.	N/A	N/A
2004/0171930	12/2003	Grimm et al.	N/A	N/A
2004/0238732	12/2003	State et al.	N/A	N/A
2005/0017972	12/2004	Poole et al.	N/A	N/A
2005/0024586	12/2004	Teiwes et al.	N/A	N/A
2005/0119639	12/2004	McCombs et al.	N/A	N/A
2005/0154296	12/2004	Lechner et al.	N/A	N/A
2005/0203367	12/2004	Ahmed et al.	N/A	N/A
2005/0203380	12/2004	Sauer et al.	N/A	N/A
2005/0215879	12/2004	Chuanggui	N/A	N/A
2005/0267358	12/2004	Tuma et al.	N/A	N/A
2006/0072124	12/2005	Smetak et al.	N/A	N/A

2006/0134198	12/2005	Tawa et al.	N/A	N/A
2006/0147100	12/2005	Fitzpatrick	N/A	N/A
2006/0176242	12/2005	Jaramaz et al.	N/A	N/A
2006/0241760	12/2005	Randall et al.	N/A	N/A
2007/0018975	12/2006	Chuanggui et al.	N/A	N/A
2007/0058261	12/2006	Sugihara et al.	N/A	N/A
2007/0100325	12/2006	Jutras et al.	N/A	N/A
2007/0183041	12/2006	McCloy et al.	N/A	N/A
2007/0233371	12/2006	Stoschek et al.	N/A	N/A
2007/0273610	12/2006	Baillot	N/A	N/A
2008/0002809	12/2007	Bodduluri	N/A	N/A
2008/0007645	12/2007	McCutchen	N/A	N/A
2008/0035266	12/2007	Danziger	N/A	N/A
2008/0085033	12/2007	Haven et al.	N/A	N/A
2008/0159612	12/2007	Fu et al.	N/A	N/A
2008/0183065	12/2007	Goldbach	N/A	N/A
2008/0221625	12/2007	Hufner et al.	N/A	N/A
2008/0253527	12/2007	Boyden et al.	N/A	N/A
2008/0262812	12/2007	Arata et al.	N/A	N/A
2008/0287728	12/2007	Mostafavi et al.	N/A	N/A
2009/0005961	12/2008	Grabowski et al.	N/A	N/A
2009/0018437	12/2008	Cooke	N/A	N/A
2009/0024127	12/2008	Lechner et al.	N/A	N/A
2009/0036902	12/2008	Dimaio et al.	N/A	N/A
2009/0062869	12/2008	Claverie et al.	N/A	N/A
2009/0099445	12/2008	Burger	N/A	N/A
2009/0123452	12/2008	Madison	N/A	N/A
2009/0227847	12/2008	Tepper et al.	N/A	N/A
2009/0285366	12/2008	Essenreiter et al.	N/A	N/A
2009/0300540	12/2008	Russell	N/A	N/A
2010/0076305	12/2009	Maier-Hein et al.	N/A	N/A
2010/0094308	12/2009	Tatsumi et al.	N/A	N/A
2010/0106010	12/2009	Rubner et al.	N/A	N/A
2010/0114110	12/2009	Taft et al.	N/A	N/A
2010/0138939	12/2009	Bentzon et al.	N/A	N/A
2010/0149073	12/2009	Chaum et al.	N/A	N/A
2010/0172567	12/2009	Prokoski	N/A	N/A
2010/0210939	12/2009	Hartmann et al.	N/A	N/A
2010/0266220	12/2009	Zagorchev et al.	N/A	N/A
2010/0274124	12/2009	Jascob et al.	N/A	N/A
2011/0004259	12/2010	Stallings et al.	N/A	N/A
2011/0098553	12/2010	Robbins et al.	N/A	N/A
2011/0105895	12/2010	Kornblau et al.	N/A	N/A
2011/0125159	12/2010	Hanson et al.	N/A	N/A
2011/0125160	12/2010	Bagga et al.	N/A	N/A
2011/0216060	12/2010	Weising et al.	N/A	N/A
2011/0245625	12/2010	Trovato et al.	N/A	N/A
2011/0248064	12/2010	Marczyk	N/A	N/A
2011/0254922	12/2010	Schaerer et al.	N/A	N/A
2011/0306873	12/2010	Shenai et al.	N/A	N/A

2012/0014608	12/2011	Watanabe	N/A	N/A
2012/0068913	12/2011	Bar-Zeev et al.	N/A	N/A
2012/0078236	12/2011	Schoepp	N/A	N/A
2012/0109151	12/2011	Maier-Hein et al.	N/A	N/A
2012/0143050	12/2011	Heigl	N/A	N/A
2012/0155064	12/2011	Waters	N/A	N/A
2012/0162452	12/2011	Liu	N/A	N/A
2012/0182605	12/2011	Hall et al.	N/A	N/A
2012/0201421	12/2011	Hartmann et al.	N/A	N/A
2012/0216411	12/2011	Wevers et al.	N/A	N/A
2012/0224260	12/2011	Healy et al.	N/A	N/A
2012/0238609	12/2011	Srivastava et al.	N/A	N/A
2012/0245645	12/2011	Hanson et al.	N/A	N/A
2012/0289777	12/2011	Chopra et al.	N/A	N/A
2012/0306850	12/2011	Balan et al.	N/A	N/A
2012/0320100	12/2011	Machida et al.	N/A	N/A
2013/0002928	12/2012	Imai	N/A	N/A
2013/0009853	12/2012	Hesselink et al.	N/A	N/A
2013/0038632	12/2012	Dillavou et al.	N/A	N/A
2013/0050258	12/2012	Liu et al.	N/A	N/A
2013/0050833	12/2012	Lewis et al.	N/A	N/A
2013/0057581	12/2012	Meier	N/A	N/A
2013/0079829	12/2012	Globerman et al.	N/A	N/A
2013/0083009	12/2012	Geisner et al.	N/A	N/A
2013/0106833	12/2012	Fun	N/A	N/A
2013/0135734	12/2012	Shafer et al.	N/A	N/A
2013/0135738	12/2012	Shafer et al.	N/A	N/A
2013/0190602	12/2012	Liao et al.	N/A	N/A
2013/0195338	12/2012	Xu et al.	N/A	N/A
2013/0209953	12/2012	Arlinsky et al.	N/A	N/A
2013/0212453	12/2012	Gudai et al.	N/A	N/A
2013/0234914	12/2012	Fujimaki	N/A	N/A
2013/0234935	12/2012	Griffith	N/A	N/A
2013/0237811	12/2012	Mihailescu et al.	N/A	N/A
2013/0245461	12/2012	Maier-Hein et al.	N/A	N/A
2013/0249787	12/2012	Morimoto	N/A	N/A
2013/0249945	12/2012	Kobayashi	N/A	N/A
2013/0265623	12/2012	Sugiyama et al.	N/A	N/A
2013/0267838	12/2012	Fronk et al.	N/A	N/A
2013/0278631	12/2012	Border et al.	N/A	N/A
2013/0278635	12/2012	Maggiore	N/A	N/A
2013/0300637	12/2012	Smits et al.	N/A	N/A
2013/0300760	12/2012	Sugano et al.	N/A	N/A
2013/0342571	12/2012	Kinnebrew et al.	N/A	N/A
2013/0345718	12/2012	Crawford et al.	N/A	N/A
2014/0031668	12/2013	Mobasser et al.	N/A	N/A
2014/0049629	12/2013	Siewerdsen et al.	N/A	N/A
2014/0088402	12/2013	Xu	N/A	N/A
2014/0088990	12/2013	Nawana et al.	N/A	N/A
2014/0104505	12/2013	Koenig	N/A	N/A

2014/0105912	12/2013	Noelle	N/A	N/A
2014/0114173	12/2013	Bar-Tal et al.	N/A	N/A
2014/0142426	12/2013	Razzaque et al.	N/A	N/A
2014/0168261	12/2013	Margolis et al.	N/A	N/A
2014/0176661	12/2013	Smurro et al.	N/A	N/A
2014/0177023	12/2013	Gao et al.	N/A	N/A
2014/0189508	12/2013	Granchi et al.	N/A	N/A
2014/0198129	12/2013	Liu et al.	N/A	N/A
2014/0218291	12/2013	Kirk	N/A	N/A
2014/0240484	12/2013	Kodama et al.	N/A	N/A
2014/0243614	12/2013	Rothberg et al.	N/A	N/A
2014/0256429	12/2013	Kobayashi et al.	N/A	N/A
2014/0266983	12/2013	Christensen	N/A	N/A
2014/0268356	12/2013	Bolas et al.	N/A	N/A
2014/0270505	12/2013	McCarthy	N/A	N/A
2014/0275760	12/2013	Lee et al.	N/A	N/A
2014/0285404	12/2013	Takano et al.	N/A	N/A
2014/0285429	12/2013	Simmons	N/A	N/A
2014/0288413	12/2013	Hwang et al.	N/A	N/A
2014/0300632	12/2013	Laor	N/A	N/A
2014/0300967	12/2013	Tilleman et al.	N/A	N/A
2014/0301624	12/2013	Barckow et al.	N/A	N/A
2014/0303491	12/2013	Shekhar et al.	N/A	N/A
2014/0320399	12/2013	Kim et al.	N/A	N/A
2014/0333899	12/2013	Smithwick	N/A	N/A
2014/0336461	12/2013	Reiter et al.	N/A	N/A
2014/0340286	12/2013	Machida et al.	N/A	N/A
2014/0361956	12/2013	Mikhailov et al.	N/A	N/A
2014/0371728	12/2013	Vaughn	N/A	N/A
2015/0005772	12/2014	Anglin et al.	N/A	N/A
2015/0018672	12/2014	Blumhofer et al.	N/A	N/A
2015/0031985	12/2014	Reddy et al.	N/A	N/A
2015/0043798	12/2014	Carrell et al.	N/A	N/A
2015/0070347	12/2014	Hofmann et al.	N/A	N/A
2015/0084990	12/2014	Laor	N/A	N/A
2015/0148847	12/2014	Moskowitz et al.	N/A	N/A
2015/0150641	12/2014	Daon et al.	N/A	N/A
2015/0182293	12/2014	Yang et al.	N/A	N/A
2015/0192776	12/2014	Lee et al.	N/A	N/A
2015/0209119	12/2014	Theodore et al.	N/A	N/A
2015/0230873	12/2014	Kubiak et al.	N/A	N/A
2015/0230893	12/2014	Huwais	N/A	N/A
2015/0261922	12/2014	Nawana et al.	N/A	N/A
2015/0277123	12/2014	Chaum et al.	N/A	N/A
2015/0282735	12/2014	Rossner	N/A	N/A
2015/0287188	12/2014	Gazit et al.	N/A	N/A
2015/0287236	12/2014	Winne et al.	N/A	N/A
2015/0297314	12/2014	Fowler et al.	N/A	N/A
2015/0305828	12/2014	Park et al.	N/A	N/A
2015/0310668	12/2014	Ellerbrock	N/A	N/A

2015/0338652	12/2014	Lim et al.	N/A	N/A
2015/0338653	12/2014	Subramaniam et al.	N/A	N/A
2015/0350517	12/2014	Duret et al.	N/A	N/A
2015/0351863	12/2014	Plassky et al.	N/A	N/A
2015/0363978	12/2014	Maimone et al.	N/A	N/A
2015/0366620	12/2014	Cameron et al.	N/A	N/A
2016/0015878	12/2015	Graham et al.	N/A	N/A
2016/0022287	12/2015	Nehls	N/A	N/A
2016/0030131	12/2015	Yang et al.	N/A	N/A
2016/0054571	12/2015	Tazbaz et al.	N/A	N/A
2016/0086380	12/2015	Vayser et al.	N/A	N/A
2016/0103318	12/2015	Du et al.	N/A	N/A
2016/0125603	12/2015	Tanji	N/A	N/A
2016/0133051	12/2015	Aonuma et al.	N/A	N/A
2016/0143699	12/2015	Tanji	N/A	N/A
2016/0153004	12/2015	Zhang et al.	N/A	N/A
2016/0163045	12/2015	Penney et al.	N/A	N/A
2016/0175064	12/2015	Steinle et al.	N/A	N/A
2016/0178910	12/2015	Giudicelli et al.	N/A	N/A
2016/0191887	12/2015	Casas	348/47	G02B 27/0172
2016/0223822	12/2015	Harrison et al.	N/A	N/A
2016/0228033	12/2015	Rossner	N/A	N/A
2016/0246059	12/2015	Halpin et al.	N/A	N/A
2016/0249989	12/2015	Devam et al.	N/A	N/A
2016/0256223	12/2015	Haimerl et al.	N/A	N/A
2016/0275684	12/2015	Elenbaas et al.	N/A	N/A
2016/0297315	12/2015	Gonzalez et al.	N/A	N/A
2016/0302870	12/2015	Wilkinson et al.	N/A	N/A
2016/0324580	12/2015	Esterberg	N/A	N/A
2016/0324583	12/2015	Kheradpir et al.	N/A	N/A
2016/0339337	12/2015	Ellsworth et al.	N/A	N/A
2017/0014119	12/2016	Capote et al.	N/A	N/A
2017/0024634	12/2016	Miao et al.	N/A	N/A
2017/0027650	12/2016	Merck et al.	N/A	N/A
2017/0031163	12/2016	Gao et al.	N/A	N/A
2017/0031179	12/2016	Guillot et al.	N/A	N/A
2017/0045742	12/2016	Greenhalgh et al.	N/A	N/A
2017/0065364	12/2016	Schuh et al.	N/A	N/A
2017/0068119	12/2016	Antaki et al.	N/A	N/A
2017/0076501	12/2016	Jagga et al.	N/A	N/A
2017/0086941	12/2016	Marti et al.	N/A	N/A
2017/0112586	12/2016	Dhupar	N/A	N/A
2017/0148215	12/2016	Aksoy et al.	N/A	N/A
2017/0164919	12/2016	Lavallee et al.	N/A	N/A
2017/0164920	12/2016	Lavallee et al.	N/A	N/A
2017/0172755	12/2016	Suh et al.	N/A	N/A
2017/0178375	12/2016	Benishti et al.	N/A	N/A
2017/0220224	12/2016	Kodali et al.	N/A	N/A
2017/0239015	12/2016	Sela et al.	N/A	N/A

2017/0245944	12/2016	Crawford et al.	N/A	N/A
2017/0251900	12/2016	Hansen et al.	N/A	N/A
2017/0252109	12/2016	Yang et al.	N/A	N/A
2017/0258526	12/2016	Lang	N/A	N/A
2017/0281283	12/2016	Siegler et al.	N/A	N/A
2017/0312032	12/2016	Amanatullah et al.	N/A	N/A
2017/0318235	12/2016	Schneider et al.	N/A	N/A
2017/0322950	12/2016	Han et al.	N/A	N/A
2017/0348055	12/2016	Salcedo et al.	N/A	N/A
2017/0348061	12/2016	Joshi et al.	N/A	N/A
2017/0366773	12/2016	Kiraly et al.	N/A	N/A
2017/0367766	12/2016	Mahfouz	N/A	N/A
2017/0367771	12/2016	Tako et al.	N/A	N/A
2017/0372477	12/2016	Penney et al.	N/A	N/A
2018/0003981	12/2017	Urey	N/A	N/A
2018/0018791	12/2017	Guoyi	N/A	N/A
2018/0021597	12/2017	Berlinger et al.	N/A	N/A
2018/0028266	12/2017	Barnes et al.	N/A	N/A
2018/0036884	12/2017	Chen et al.	N/A	N/A
2018/0049622	12/2017	Ryan et al.	N/A	N/A
2018/0055579	12/2017	Daon et al.	N/A	N/A
2018/0071029	12/2017	Srimohanarajah et al.	N/A	N/A
2018/0078316	12/2017	Schaewe et al.	N/A	N/A
2018/0082480	12/2017	White et al.	N/A	N/A
2018/0092667	12/2017	Heigl et al.	N/A	N/A
2018/0092698	12/2017	Chopra et al.	N/A	N/A
2018/0092699	12/2017	Finley	N/A	N/A
2018/0116732	12/2017	Lin et al.	N/A	N/A
2018/0116741	12/2017	Garcia et al.	N/A	N/A
2018/0117150	12/2017	O'Dwyer et al.	N/A	N/A
2018/0120106	12/2017	Sato	N/A	N/A
2018/0133871	12/2017	Farmer	N/A	N/A
2018/0143442	12/2017	Gupta	N/A	N/A
2018/0153626	12/2017	Yang et al.	N/A	N/A
2018/0182150	12/2017	Benishti et al.	N/A	N/A
2018/0185100	12/2017	Weinstein et al.	N/A	N/A
2018/0185113	12/2017	Gregerson et al.	N/A	N/A
2018/0193097	12/2017	McLachlin et al.	N/A	N/A
2018/0200002	12/2017	Kostrzewski et al.	N/A	N/A
2018/0247128	12/2017	Alvi et al.	N/A	N/A
2018/0262743	12/2017	Casas	N/A	N/A
2018/0303558	12/2017	Thomas	N/A	N/A
2018/0311011	12/2017	Van et al.	N/A	N/A
2018/0317803	12/2017	Ben-Yishai et al.	N/A	N/A
2018/0318035	12/2017	McLachlin et al.	N/A	N/A
2018/0368898	12/2017	Divincenzo et al.	N/A	N/A
2019/0000372	12/2018	Gullotti et al.	N/A	N/A
2019/0000564	12/2018	Navab et al.	N/A	N/A
2019/0015163	12/2018	Abhari et al.	N/A	N/A
2019/0018235	12/2018	Ouderkirk et al.	N/A	N/A

2019/0038362	12/2018	Nash et al.	N/A	N/A
2019/0038365	12/2018	Soper et al.	N/A	N/A
2019/0043238	12/2018	Benishti et al.	N/A	N/A
2019/0043392	12/2018	Abele	N/A	N/A
2019/0046272	12/2018	Zoabi et al.	N/A	N/A
2019/0046276	12/2018	Inglese et al.	N/A	N/A
2019/0053851	12/2018	Siemionow et al.	N/A	N/A
2019/0069971	12/2018	Tripathi et al.	N/A	N/A
2019/0080515	12/2018	Geri et al.	N/A	N/A
2019/0105116	12/2018	Johnson et al.	N/A	N/A
2019/0130792	12/2018	Rios et al.	N/A	N/A
2019/0142519	12/2018	Siemionow et al.	N/A	N/A
2019/0144443	12/2018	Jackson et al.	N/A	N/A
2019/0175228	12/2018	Elimelech et al.	N/A	N/A
2019/0192226	12/2018	Lang	N/A	N/A
2019/0192230	12/2018	Siemionow et al.	N/A	N/A
2019/0200894	12/2018	Jung et al.	N/A	N/A
2019/0201106	12/2018	Siemionow et al.	N/A	N/A
2019/0205606	12/2018	Zhou et al.	N/A	N/A
2019/0216537	12/2018	Eltorai et al.	N/A	N/A
2019/0251692	12/2018	Schmidt-Richberg et al.	N/A	N/A
2019/0251694	12/2018	Han et al.	N/A	N/A
2019/0254753	12/2018	Johnson et al.	N/A	N/A
2019/0273916	12/2018	Benishti et al.	N/A	N/A
2019/0310481	12/2018	Blum et al.	N/A	N/A
2019/0324365	12/2018	De et al.	N/A	N/A
2019/0333480	12/2018	Lang	N/A	N/A
2019/0369660	12/2018	Wen et al.	N/A	N/A
2019/0369717	12/2018	Frielinghaus et al.	N/A	N/A
2019/0378276	12/2018	Flossmann et al.	N/A	N/A
2019/0387351	12/2018	Lyren et al.	N/A	N/A
2020/0015895	12/2019	Frielinghaus et al.	N/A	N/A
2020/0019364	12/2019	Pond	N/A	N/A
2020/0020249	12/2019	Jarc et al.	N/A	N/A
2020/0038112	12/2019	Amanatullah et al.	N/A	N/A
2020/0043160	12/2019	Mizukura et al.	N/A	N/A
2020/0059640	12/2019	Browd et al.	N/A	N/A
2020/0078100	12/2019	Weinstein et al.	N/A	N/A
2020/0085511	12/2019	Oezbek et al.	N/A	N/A
2020/0088997	12/2019	Lee et al.	N/A	N/A
2020/0100847	12/2019	Siegler et al.	N/A	N/A
2020/0117025	12/2019	Sauer	N/A	N/A
2020/0129058	12/2019	Li et al.	N/A	N/A
2020/0129136	12/2019	Harding et al.	N/A	N/A
2020/0129262	12/2019	Verard et al.	N/A	N/A
2020/0129264	12/2019	Oativia et al.	N/A	N/A
2020/0133029	12/2019	Yonezawa	N/A	N/A
2020/0138518	12/2019	Lang	N/A	N/A
2020/0138618	12/2019	Roszkowiak et al.	N/A	N/A

2020/0143594	12/2019	Lal et al.	N/A	N/A
2020/0146546	12/2019	Chene et al.	N/A	N/A
2020/0151507	12/2019	Siemionow et al.	N/A	N/A
2020/0156259	12/2019	Ruiz et al.	N/A	N/A
2020/0159313	12/2019	Gibby et al.	N/A	N/A
2020/0163723	12/2019	Wolf et al.	N/A	N/A
2020/0163739	12/2019	Messinger et al.	N/A	N/A
2020/0178916	12/2019	Lalys et al.	N/A	N/A
2020/0184638	12/2019	Meglan et al.	N/A	N/A
2020/0186786	12/2019	Gibby et al.	N/A	N/A
2020/0188028	12/2019	Feiner et al.	N/A	N/A
2020/0188034	12/2019	Lequette et al.	N/A	N/A
2020/0201082	12/2019	Carabin	N/A	N/A
2020/0229877	12/2019	Siemionow et al.	N/A	N/A
2020/0237256	12/2019	Farshad et al.	N/A	N/A
2020/0237459	12/2019	Racheli et al.	N/A	N/A
2020/0237880	12/2019	Kent et al.	N/A	N/A
2020/0242280	12/2019	Pavloff et al.	N/A	N/A
2020/0246074	12/2019	Lang	N/A	N/A
2020/0246081	12/2019	Johnson et al.	N/A	N/A
2020/0264451	12/2019	Blum et al.	N/A	N/A
2020/0265273	12/2019	Wei et al.	N/A	N/A
2020/0275988	12/2019	Johnson et al.	N/A	N/A
2020/0281554	12/2019	Trini et al.	N/A	N/A
2020/0286222	12/2019	Essenreiter et al.	N/A	N/A
2020/0288075	12/2019	Bonin et al.	N/A	N/A
2020/0294233	12/2019	Merlet	N/A	N/A
2020/0297427	12/2019	Cameron et al.	N/A	N/A
2020/0305980	12/2019	Lang	N/A	N/A
2020/0315734	12/2019	El Amm	N/A	N/A
2020/0321099	12/2019	Holladay et al.	N/A	N/A
2020/0323460	12/2019	Busza et al.	N/A	N/A
2020/0323609	12/2019	Johnson et al.	N/A	N/A
2020/0327721	12/2019	Siemionow et al.	N/A	N/A
2020/0330179	12/2019	Ton	N/A	N/A
2020/0337780	12/2019	Winkler et al.	N/A	N/A
2020/0341283	12/2019	McCracken et al.	N/A	N/A
2020/0352655	12/2019	Freese	N/A	N/A
2020/0355927	12/2019	Marcellin-Dibon et al.	N/A	N/A
2020/0360091	12/2019	Murray et al.	N/A	N/A
2020/0360105	12/2019	Frey et al.	N/A	N/A
2020/0375666	12/2019	Murphy	N/A	N/A
2020/0377493	12/2019	Heiser et al.	N/A	N/A
2020/0377956	12/2019	Vogelstein et al.	N/A	N/A
2020/0386982	12/2019	Luxembourg	N/A	N/A
2020/0388075	12/2019	Kazanzides et al.	N/A	N/A
2020/0389425	12/2019	Bhatia et al.	N/A	N/A
2020/0390502	12/2019	Holthuizen et al.	N/A	N/A
2020/0390503	12/2019	Casas et al.	N/A	N/A
2020/0402647	12/2019	Domracheva et al.	N/A	N/A

2020/0409306	12/2019	Gelman et al.	N/A	N/A
2020/0410687	12/2019	Siemionow et al.	N/A	N/A
2020/0413031	12/2019	Khani et al.	N/A	N/A
2021/0004956	12/2020	Book et al.	N/A	N/A
2021/0009339	12/2020	Morrison et al.	N/A	N/A
2021/0015560	12/2020	Boddington et al.	N/A	N/A
2021/0015583	12/2020	Avisar et al.	N/A	N/A
2021/0022599	12/2020	Freeman et al.	N/A	N/A
2021/0022808	12/2020	Lang	N/A	N/A
2021/0022811	12/2020	Mahfouz	N/A	N/A
2021/0022828	12/2020	Elimelech et al.	N/A	N/A
2021/0029804	12/2020	Chang	N/A	N/A
2021/0030374	12/2020	Takahashi et al.	N/A	N/A
2021/0030511	12/2020	Wolf et al.	N/A	N/A
2021/0038339	12/2020	Yu et al.	N/A	N/A
2021/0049825	12/2020	Wheelwright et al.	N/A	N/A
2021/0052348	12/2020	Stifter et al.	N/A	N/A
2021/0056687	12/2020	Hibbard et al.	N/A	N/A
2021/0065911	12/2020	Goel et al.	N/A	N/A
2021/0077195	12/2020	Saeidi et al.	N/A	N/A
2021/0077210	12/2020	Itkowitz et al.	N/A	N/A
2021/0080751	12/2020	Lindsey et al.	N/A	N/A
2021/0090344	12/2020	Geri et al.	N/A	N/A
2021/0093391	12/2020	Poltaretskyi et al.	N/A	N/A
2021/0093392	12/2020	Poltaretskyi et al.	N/A	N/A
2021/0093400	12/2020	Quaid et al.	N/A	N/A
2021/0093417	12/2020	Liu	N/A	N/A
2021/0104055	12/2020	Ni et al.	N/A	N/A
2021/0107923	12/2020	Jackson et al.	N/A	N/A
2021/0109349	12/2020	Schneider et al.	N/A	N/A
2021/0109373	12/2020	Loo et al.	N/A	N/A
2021/0110517	12/2020	Flohr et al.	N/A	N/A
2021/0113269	12/2020	Vilsmeier et al.	N/A	N/A
2021/0113293	12/2020	Silva et al.	N/A	N/A
2021/0121238	12/2020	Palushi et al.	N/A	N/A
2021/0137634	12/2020	Lang	N/A	N/A
2021/0141887	12/2020	Kim et al.	N/A	N/A
2021/0150702	12/2020	Claessen et al.	N/A	N/A
2021/0157544	12/2020	Denton	N/A	N/A
2021/0160472	12/2020	Casas	N/A	N/A
2021/0161614	12/2020	Elimelech et al.	N/A	N/A
2021/0162287	12/2020	Xing et al.	N/A	N/A
2021/0165207	12/2020	Peyman	N/A	N/A
2021/0169504	12/2020	Brown	N/A	N/A
2021/0169578	12/2020	Calloway et al.	N/A	N/A
2021/0169581	12/2020	Calloway et al.	N/A	N/A
2021/0169605	12/2020	Calloway et al.	N/A	N/A
2021/0186647	12/2020	Elimelech et al.	N/A	N/A
2021/0196404	12/2020	Wang	N/A	N/A
2021/0211640	12/2020	Bristol et al.	N/A	N/A

2021/0223577	12/2020	Zhang et al.	N/A	N/A
2021/0225006	12/2020	Grady et al.	N/A	N/A
2021/0227791	12/2020	De et al.	N/A	N/A
2021/0231301	12/2020	Hikmet et al.	N/A	N/A
2021/0235061	12/2020	Hegyi	N/A	N/A
2021/0248822	12/2020	Choi et al.	N/A	N/A
2021/0274281	12/2020	Zhang et al.	N/A	N/A
2021/0278675	12/2020	Klug et al.	N/A	N/A
2021/0282887	12/2020	Wiggenermann	N/A	N/A
2021/0290046	12/2020	Nazareth et al.	N/A	N/A
2021/0290336	12/2020	Wang	N/A	N/A
2021/0290394	12/2020	Mahfouz	N/A	N/A
2021/0295108	12/2020	Bar	N/A	N/A
2021/0295512	12/2020	Knoplioch et al.	N/A	N/A
2021/0298795	12/2020	Bowling et al.	N/A	N/A
2021/0298835	12/2020	Wang	N/A	N/A
2021/0306599	12/2020	Pierce	N/A	N/A
2021/0311322	12/2020	Belanger et al.	N/A	N/A
2021/0314502	12/2020	Liu	N/A	N/A
2021/0315636	12/2020	Akbarian et al.	N/A	N/A
2021/0315662	12/2020	Freeman et al.	N/A	N/A
2021/0325684	12/2020	Ninan et al.	N/A	N/A
2021/0332447	12/2020	Lubelski et al.	N/A	N/A
2021/0333561	12/2020	Oh et al.	N/A	N/A
2021/0341739	12/2020	Cakmakci et al.	N/A	N/A
2021/0341740	12/2020	Cakmakci et al.	N/A	N/A
2021/0346115	12/2020	Dulin et al.	N/A	N/A
2021/0349677	12/2020	Baldev et al.	N/A	N/A
2021/0364802	12/2020	Uchiyama et al.	N/A	N/A
2021/0369226	12/2020	Siemionow et al.	N/A	N/A
2021/0371413	12/2020	Thurston et al.	N/A	N/A
2021/0373333	12/2020	Moon	N/A	N/A
2021/0373344	12/2020	Loyola et al.	N/A	N/A
2021/0378757	12/2020	Bay et al.	N/A	N/A
2021/0382310	12/2020	Freeman et al.	N/A	N/A
2021/0386482	12/2020	Gera et al.	N/A	N/A
2021/0389590	12/2020	Freeman et al.	N/A	N/A
2021/0400247	12/2020	Casas	N/A	N/A
2021/0401533	12/2020	Im	N/A	N/A
2021/0402255	12/2020	Fung	N/A	N/A
2021/0405369	12/2020	King	N/A	N/A
2022/0003992	12/2021	Ahn	N/A	N/A
2022/0007006	12/2021	Healy et al.	N/A	N/A
2022/0008135	12/2021	Frielinghaus et al.	N/A	N/A
2022/0038675	12/2021	Hegyi	N/A	N/A
2022/0039873	12/2021	Harris	N/A	N/A
2022/0051484	12/2021	Jones et al.	N/A	N/A
2022/0054199	12/2021	Sivaprakasam et al.	N/A	N/A
2022/0061921	12/2021	Crawford et al.	N/A	N/A
2022/0071712	12/2021	Wolf et al.	N/A	N/A

2022/0079675	12/2021	Lang	N/A	N/A
2022/0087746	12/2021	Lang	N/A	N/A
2022/0113810	12/2021	Isaacs et al.	N/A	N/A
2022/0117669	12/2021	Nikou et al.	N/A	N/A
2022/0121041	12/2021	Hakim	N/A	N/A
2022/0125496	12/2021	Lpez et al.	N/A	N/A
2022/0133484	12/2021	Lang	N/A	N/A
2022/0142730	12/2021	Wolf et al.	N/A	N/A
2022/0155861	12/2021	Myung et al.	N/A	N/A
2022/0159227	12/2021	Quiles Casas	N/A	N/A
2022/0179209	12/2021	Cherukuri	N/A	N/A
2022/0192776	12/2021	Gibby et al.	N/A	N/A
2022/0193453	12/2021	Miyazaki et al.	N/A	N/A
2022/0201274	12/2021	Achilefu et al.	N/A	N/A
2022/0245400	12/2021	Siemionow et al.	N/A	N/A
2022/0245821	12/2021	Ouzounis	N/A	G06T 7/62
2022/0257206	12/2021	Hartley et al.	N/A	N/A
2022/0269077	12/2021	Adema et al.	N/A	N/A
2022/0270263	12/2021	Junio	N/A	N/A
2022/0287676	12/2021	Steines et al.	N/A	N/A
2022/0292786	12/2021	Pelzl et al.	N/A	N/A
2022/0295033	12/2021	Quiles Casas	N/A	N/A
2022/0296315	12/2021	Sokhanvar et al.	N/A	N/A
2022/0304768	12/2021	Elimelech et al.	N/A	N/A
2022/0351385	12/2021	Finley et al.	N/A	N/A
2022/0353487	12/2021	Hegyí	N/A	N/A
2022/0358759	12/2021	Cork et al.	N/A	N/A
2022/0370152	12/2021	Lavallee et al.	N/A	N/A
2022/0387130	12/2021	Spaas et al.	N/A	N/A
2022/0392085	12/2021	Finley et al.	N/A	N/A
2022/0397750	12/2021	Zhou et al.	N/A	N/A
2022/0398752	12/2021	Yoon et al.	N/A	N/A
2022/0398755	12/2021	Herrmann	N/A	N/A
2022/0405935	12/2021	Flossmann et al.	N/A	N/A
2023/0004013	12/2022	Mccracken et al.	N/A	N/A
2023/0009793	12/2022	Gera et al.	N/A	N/A
2023/0025480	12/2022	Kemp et al.	N/A	N/A
2023/0027801	12/2022	Qian et al.	N/A	N/A
2023/0032731	12/2022	Hrndler et al.	N/A	N/A
2023/0034189	12/2022	Gera et al.	N/A	N/A
2023/0050636	12/2022	Yanof et al.	N/A	N/A
2023/0053120	12/2022	Jamali et al.	N/A	N/A
2023/0073041	12/2022	Samadani et al.	N/A	N/A
2023/0085387	12/2022	Jones et al.	N/A	N/A
2023/0087783	12/2022	Dulin et al.	N/A	N/A
2023/0100078	12/2022	Toporek et al.	N/A	N/A
2023/0123621	12/2022	Joshi et al.	N/A	N/A
2023/0126207	12/2022	Wang	N/A	N/A
2023/0129056	12/2022	Hemingway et al.	N/A	N/A
2023/0131515	12/2022	Oezbek et al.	N/A	N/A

2023/0149083	12/2022	Lin et al.	N/A	N/A
2023/0162493	12/2022	Worrell et al.	N/A	N/A
2023/0165640	12/2022	Dulin et al.	N/A	N/A
2023/0169659	12/2022	Chen et al.	N/A	N/A
2023/0196582	12/2022	Grady et al.	N/A	N/A
2023/0200917	12/2022	Calloway et al.	N/A	N/A
2023/0236426	12/2022	Manly et al.	N/A	N/A
2023/0236427	12/2022	Jiannyuh	N/A	N/A
2023/0245784	12/2022	Crawford et al.	N/A	N/A
2023/0260142	12/2022	Chatterjee et al.	N/A	N/A
2023/0290037	12/2022	Tasse et al.	N/A	N/A
2023/0295302	12/2022	Bhagavatheeswaran et al.	N/A	N/A
2023/0306590	12/2022	Jazdzyk et al.	N/A	N/A
2023/0316550	12/2022	Hiasa	N/A	N/A
2023/0326011	12/2022	Cutforth et al.	N/A	N/A
2023/0326027	12/2022	Wahrenberg	N/A	N/A
2023/0329799	12/2022	Gera et al.	N/A	N/A
2023/0329801	12/2022	Elimelech et al.	N/A	N/A
2023/0334664	12/2022	Lu et al.	N/A	N/A
2023/0335261	12/2022	Reicher et al.	N/A	N/A
2023/0359043	12/2022	Russell	N/A	N/A
2023/0363832	12/2022	Mosadegh et al.	N/A	N/A
2023/0371984	12/2022	Leuthardt et al.	N/A	N/A
2023/0372053	12/2022	Elimelech et al.	N/A	N/A
2023/0372054	12/2022	Elimelech et al.	N/A	N/A
2023/0377171	12/2022	Hasler et al.	N/A	N/A
2023/0377175	12/2022	Seok	N/A	N/A
2023/0379448	12/2022	Benishti et al.	N/A	N/A
2023/0379449	12/2022	Benishti et al.	N/A	N/A
2023/0386022	12/2022	Tan et al.	N/A	N/A
2023/0386067	12/2022	De et al.	N/A	N/A
2023/0389991	12/2022	Glaser et al.	N/A	N/A
2023/0394791	12/2022	Wang et al.	N/A	N/A
2023/0397349	12/2022	Capelli et al.	N/A	N/A
2023/0397957	12/2022	Crawford et al.	N/A	N/A
2023/0410445	12/2022	Elimelech et al.	N/A	N/A
2023/0419496	12/2022	Wuelker et al.	N/A	N/A
2023/0420114	12/2022	Scholler et al.	N/A	N/A
2024/0008935	12/2023	Wolf et al.	N/A	N/A
2024/0016549	12/2023	Johnson et al.	N/A	N/A
2024/0016572	12/2023	Elimelech et al.	N/A	N/A
2024/0020831	12/2023	Johnson et al.	N/A	N/A
2024/0020840	12/2023	Johnson et al.	N/A	N/A
2024/0020862	12/2023	Johnson et al.	N/A	N/A
2024/0022704	12/2023	Benishti et al.	N/A	N/A
2024/0023946	12/2023	Wolf et al.	N/A	N/A
2024/0041530	12/2023	Lang	N/A	N/A
2024/0041558	12/2023	Siewerdsen et al.	N/A	N/A
2024/0045491	12/2023	Sourov	N/A	N/A

2024/0058064	12/2023	Weiser et al.	N/A	N/A
2024/0062387	12/2023	Frantz et al.	N/A	N/A
2024/0103271	12/2023	Zare Seisan	N/A	N/A
2024/0103282	12/2023	Law et al.	N/A	N/A
2024/0111163	12/2023	Law et al.	N/A	N/A
2024/0122560	12/2023	Junio et al.	N/A	N/A
2024/0126087	12/2023	Gera et al.	N/A	N/A
2024/0127578	12/2023	Hiasa	N/A	N/A
2024/0129451	12/2023	Healy et al.	N/A	N/A
2024/0130826	12/2023	Elimelech et al.	N/A	N/A
2024/0134206	12/2023	Gera et al.	N/A	N/A
2024/0144497	12/2023	Cvetko et al.	N/A	N/A
2024/0156532	12/2023	Weiman et al.	N/A	N/A
2024/0177445	12/2023	Galeotti et al.	N/A	N/A
2024/0177458	12/2023	Zhang et al.	N/A	N/A
2024/0180634	12/2023	Mikus	N/A	N/A
2024/0184119	12/2023	Lee et al.	N/A	N/A
2024/0185509	12/2023	Kovler et al.	N/A	N/A
2024/0202926	12/2023	Crawford et al.	N/A	N/A
2024/0202927	12/2023	Haslam et al.	N/A	N/A
2024/0212111	12/2023	Genghi et al.	N/A	N/A
2024/0233131	12/2023	Westerhoff et al.	N/A	N/A
2024/0245463	12/2023	Mlsmeier et al.	N/A	N/A
2024/0245474	12/2023	Weiman et al.	N/A	N/A
2024/0248530	12/2023	Gibby et al.	N/A	N/A
2024/0252252	12/2023	Lang	N/A	N/A
2024/0261036	12/2023	Finley et al.	N/A	N/A
2024/0261058	12/2023	Gera et al.	N/A	N/A
2024/0265645	12/2023	Papar	N/A	N/A
2024/0266033	12/2023	Freeman et al.	N/A	N/A
2024/0268922	12/2023	Calloway et al.	N/A	N/A
2024/0273740	12/2023	Gibby et al.	N/A	N/A
2024/0281979	12/2023	Schrempf et al.	N/A	N/A
2024/0296527	12/2023	Nett et al.	N/A	N/A
2024/0303832	12/2023	Chen et al.	N/A	N/A
2024/0307101	12/2023	Gera et al.	N/A	N/A
2024/0312012	12/2023	Li et al.	N/A	N/A
2024/0341853	12/2023	Gibby et al.	N/A	N/A
2024/0341861	12/2023	Wolf et al.	N/A	N/A
2024/0341910	12/2023	Wolf et al.	N/A	N/A
2024/0341911	12/2023	Elimelech et al.	N/A	N/A
2024/0355098	12/2023	Liu et al.	N/A	N/A
2024/0374314	12/2023	Frey et al.	N/A	N/A
2024/0377640	12/2023	Asaban et al.	N/A	N/A
2024/0378708	12/2023	Kim et al.	N/A	N/A
2024/0382283	12/2023	Kuhnert et al.	N/A	N/A
2024/0386572	12/2023	Barasofsky et al.	N/A	N/A
2024/0386682	12/2023	Cvetko et al.	N/A	N/A
2024/0394883	12/2023	Liao et al.	N/A	N/A
2024/0394985	12/2023	Hanlon et al.	N/A	N/A

2024/0404065	12/2023	Gibbons et al.	N/A	N/A
2024/0404106	12/2023	Wu et al.	N/A	N/A
2024/0404180	12/2023	Kobayashi et al.	N/A	N/A
2024/0420337	12/2023	Li et al.	N/A	N/A
2024/0420592	12/2023	Stone et al.	N/A	N/A
2024/0423724	12/2023	Wolf et al.	N/A	N/A
2024/0423750	12/2023	Elimelech et al.	N/A	N/A
2025/0020931	12/2024	Gera et al.	N/A	N/A
2025/0049534	12/2024	Elimelech et al.	N/A	N/A
2025/0090266	12/2024	Gera et al.	N/A	N/A
2025/0114151	12/2024	Gera et al.	N/A	N/A
2025/0114164	12/2024	Gera et al.	N/A	N/A
2025/0114165	12/2024	Wolf et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3022448	12/2017	CA	N/A
3034314	12/2017	CA	N/A
101379412	12/2008	CN	N/A
102740784	12/2011	CN	N/A
102740789	12/2011	CN	N/A
103106348	12/2012	CN	N/A
103945780	12/2013	CN	N/A
105310756	12/2015	CN	N/A
109199563	12/2018	CN	N/A
111915696	12/2019	CN	N/A
112489047	12/2020	CN	N/A
202004011567	12/2003	DE	N/A
102004011567	12/2004	DE	N/A
102014008153	12/2013	DE	N/A
202022103168	12/2021	DE	N/A
0933096	12/1998	EP	N/A
1640750	12/2005	EP	N/A
1757974	12/2006	EP	N/A
2119397	12/2008	EP	N/A
2134847	12/2008	EP	N/A
2557998	12/2012	EP	N/A
2823463	12/2014	EP	N/A
2868277	12/2014	EP	N/A
2891966	12/2014	EP	N/A
2963616	12/2015	EP	N/A
3028258	12/2015	EP	N/A
3034607	12/2015	EP	N/A
3037038	12/2015	EP	N/A
3069318	12/2015	EP	N/A
3076660	12/2015	EP	N/A
3121789	12/2016	EP	N/A
3123970	12/2016	EP	N/A
2654749	12/2016	EP	N/A

3175815	12/2016	EP	N/A
3216416	12/2016	EP	N/A
2032039	12/2016	EP	N/A
3224376	12/2016	EP	N/A
3247297	12/2016	EP	N/A
3256213	12/2016	EP	N/A
3306567	12/2017	EP	N/A
3320874	12/2017	EP	N/A
2030193	12/2017	EP	N/A
2225723	12/2018	EP	N/A
2619622	12/2018	EP	N/A
2892558	12/2018	EP	N/A
3494903	12/2018	EP	N/A
2635299	12/2018	EP	N/A
3505050	12/2018	EP	N/A
2875149	12/2018	EP	N/A
3593227	12/2019	EP	N/A
3634294	12/2019	EP	N/A
3206583	12/2019	EP	N/A
3711700	12/2019	EP	N/A
2625845	12/2020	EP	N/A
3789965	12/2020	EP	N/A
3858280	12/2020	EP	N/A
3913423	12/2020	EP	N/A
3952331	12/2021	EP	N/A
3960235	12/2021	EP	N/A
3635683	12/2021	EP	N/A
3602492	12/2021	EP	N/A
4173590	12/2022	EP	N/A
3533031	12/2022	EP	N/A
4252695	12/2022	EP	N/A
3195257	12/2022	EP	N/A
3405909	12/2022	EP	N/A
4270313	12/2022	EP	N/A
4287120	12/2022	EP	N/A
3488381	12/2023	EP	N/A
3834768	12/2023	EP	N/A
3903714	12/2023	EP	N/A
4336450	12/2023	EP	N/A
3814984	12/2023	EP	N/A
4115389	12/2023	EP	N/A
3752981	12/2023	EP	N/A
4375948	12/2023	EP	N/A
4383203	12/2023	EP	N/A
4459543	12/2023	EP	N/A
4292045	12/2023	EP	N/A
4298604	12/2023	EP	N/A
2507314	12/2013	GB	N/A
262864	12/2018	IL	N/A
2004-237092	12/2003	JP	N/A

2005-246059	12/2004	JP	N/A
2008-507361	12/2007	JP	N/A
2009-514571	12/2008	JP	N/A
2021-525186	12/2020	JP	N/A
10-2014-0120155	12/2013	KR	N/A
03/34705	12/2002	WO	N/A
2006/002559	12/2005	WO	N/A
2007/051304	12/2006	WO	N/A
2007/115826	12/2006	WO	N/A
2008/103383	12/2007	WO	N/A
2010/067267	12/2009	WO	N/A
2010/074747	12/2009	WO	N/A
2012/061537	12/2011	WO	N/A
2012/101286	12/2011	WO	N/A
2013/112554	12/2012	WO	N/A
2014/014498	12/2013	WO	N/A
2014/024188	12/2013	WO	N/A
2014/037953	12/2013	WO	N/A
2014/113455	12/2013	WO	N/A
2014/125789	12/2013	WO	N/A
2014/167563	12/2013	WO	N/A
2014/174067	12/2013	WO	N/A
2015/058816	12/2014	WO	N/A
2015/061752	12/2014	WO	N/A
2015/109145	12/2014	WO	N/A
2016/151506	12/2015	WO	N/A
2017/042171	12/2016	WO	N/A
2018/052966	12/2017	WO	N/A
2018/073452	12/2017	WO	N/A
2018/200767	12/2017	WO	N/A
2018/206086	12/2017	WO	N/A
2019/083431	12/2018	WO	N/A
2019/135209	12/2018	WO	N/A
2019/135210	12/2018	WO	N/A
2019/161477	12/2018	WO	N/A
2019/195926	12/2018	WO	N/A
2019/210353	12/2018	WO	N/A
2019/211741	12/2018	WO	N/A
2020/109903	12/2019	WO	N/A
2020/109904	12/2019	WO	N/A
2021/017019	12/2020	WO	N/A
2021/019369	12/2020	WO	N/A
2021/021979	12/2020	WO	N/A
2021/023574	12/2020	WO	N/A
2021/046455	12/2020	WO	N/A
2021/048158	12/2020	WO	N/A
2021/061459	12/2020	WO	N/A
2021/062375	12/2020	WO	N/A
2021/073743	12/2020	WO	N/A
2021/087439	12/2020	WO	N/A

2021/091980	12/2020	WO	N/A
2021/112918	12/2020	WO	N/A
2021/130564	12/2020	WO	N/A
2021/137752	12/2020	WO	N/A
2021/141887	12/2020	WO	N/A
2021/145584	12/2020	WO	N/A
2021/154076	12/2020	WO	N/A
2021/183318	12/2020	WO	N/A
2021/188757	12/2020	WO	N/A
2021/255627	12/2020	WO	N/A
2021/257897	12/2020	WO	N/A
2021/258078	12/2020	WO	N/A
2022/009233	12/2021	WO	N/A
2022/053923	12/2021	WO	N/A
2022/056010	12/2021	WO	N/A
2022/079565	12/2021	WO	N/A
2022/180624	12/2021	WO	N/A
2023/003952	12/2022	WO	N/A
2023/281395	12/2022	WO	N/A
2023/007418	12/2022	WO	N/A
2023/011924	12/2022	WO	N/A
2023/021448	12/2022	WO	N/A
2023/021450	12/2022	WO	N/A
2023/021451	12/2022	WO	N/A
2023/026229	12/2022	WO	N/A
2023/047355	12/2022	WO	N/A
2023/072887	12/2022	WO	N/A
2023/088986	12/2022	WO	N/A
2023/158878	12/2022	WO	N/A
2023/159104	12/2022	WO	N/A
2023/161848	12/2022	WO	N/A
2023/163933	12/2022	WO	N/A
2023/175244	12/2022	WO	N/A
2023/186996	12/2022	WO	N/A
2023/202909	12/2022	WO	N/A
2023/205212	12/2022	WO	N/A
2023/205896	12/2022	WO	N/A
2023/209014	12/2022	WO	N/A
2023/229415	12/2022	WO	N/A
2023/232492	12/2022	WO	N/A
2023/240912	12/2022	WO	N/A
2024/001140	12/2023	WO	N/A
2024/002620	12/2023	WO	N/A
2024/013642	12/2023	WO	N/A
2024/018368	12/2023	WO	N/A
2024/046760	12/2023	WO	N/A
2024/052136	12/2023	WO	N/A
2024/077077	12/2023	WO	N/A
2024/121060	12/2023	WO	N/A
2024/132609	12/2023	WO	N/A

2024/145341	12/2023	WO	N/A
2024/160896	12/2023	WO	N/A
2024/165508	12/2023	WO	N/A
2024/173251	12/2023	WO	N/A
2024/186811	12/2023	WO	N/A
2024/226797	12/2023	WO	N/A
2024/251344	12/2023	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 15/896,102 (U.S. Pat. No. 10,134,166), filed Feb. 14, 2018 (Nov. 20, 2018), Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 16/159,740 (U.S. Pat. No. 10,382,748), filed Oct. 15, 2018 (Aug. 13, 2019), Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 16/419,023 (U.S. Pat. No. 11,750,794), filed May 22, 2019 (Sep. 5, 2023), Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 18/352,158, filed Jul. 13, 2023, Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 18/365,643, filed Aug. 4, 2023, Head-Mounted Augmented Reality Near Eye Display Device. cited by applicant

U.S. Appl. No. 18/365,650, filed Aug. 4, 2023, Systems for Facilitating Augmented Reality-Assisted Medical Procedures. cited by applicant

U.S. Appl. No. 15/127,423 (U.S. Pat. No. 9,928,629), filed Sep. 20, 2016 (Mar. 27, 2018), Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 16/120,480 (U.S. Pat. No. 10,835,296), filed Sep. 4, 2018 (Nov. 17, 2020), Spinous Process Clamp. cited by applicant

U.S. Appl. No. 17/067,831, filed Oct. 12, 2020, Spinous Process Clamp. cited by applicant

U.S. Appl. No. 18/030,072, filed Apr. 4, 2023, Spinous Process Clamp. cited by applicant

U.S. Appl. No. 18/365,590 (U.S. Pat. No. 11,980,508), filed Aug. 4, 2023 (May 14, 2024), Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 18/365,571 (U.S. Pat. No. 11,974,887), filed Aug. 4, 2023 (May 7, 2024), Registration Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 18/632,588, filed Apr. 11, 2024, Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 17/045,766 (U.S. Pat. No. 11,980,507), filed Oct. 7, 2020 (May 14, 2024), Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 16/199,281 (U.S. Pat. No. 10,939,977), filed Nov. 26, 2018 (Mar. 9, 2021), Positioning Marker. cited by applicant

U.S. Appl. No. 16/524,258 (U.S. Pat. No. 11,980,506), filed Jul. 29, 2019 (May 14, 2024), Fiducial Marker. cited by applicant

U.S. Appl. No. 18/631,804, filed Apr. 10, 2024, Fiducial Marker. cited by applicant

U.S. Appl. No. 17/585,629, filed Jan. 27, 2022, Fiducial Marker. cited by applicant

U.S. Appl. No. 16/724,297 (U.S. Pat. No. 11,382,712), filed Dec. 22, 2019 (Jul. 12, 2022), Mirroring in Image Guided Surgery. cited by applicant

U.S. Appl. No. 17/827,710 (U.S. Pat. No. 11,801,115), filed May 29, 2022 (Oct. 31, 2023), Mirroring in Image Guided Surgery. cited by applicant

U.S. Appl. No. 18/352,181, filed Jul. 13, 2023, Mirroring in Image Guided Surgery. cited by

applicant
U.S. Appl. No. 18/400,739, filed Dec. 29, 2023, Mirroring in Image Guided Surgery. cited by applicant
U.S. Appl. No. 16/200,144 (U.S. Pat. No. 11,766,296), filed Nov. 26, 2018 (Sep. 26, 2023), Tracking System for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/470,809 (U.S. Pat. No. 11,980,429), filed Sep. 20, 2023 (May 14, 2024), Tracking Methods for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/631,877, filed Apr. 10, 2024, Tracking Systems and Methods for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 17/015,199, filed Sep. 9, 2020, Universal Tool Adapter. cited by applicant
U.S. Appl. No. 18/598,965, filed Mar. 7, 2024, Universal Tool Adapter for Image Guided Surgery. cited by applicant
U.S. Appl. No. 18/044,380, filed Mar. 8, 2023, Universal Tool Adapter for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 16/901,026 (U.S. Pat. No. 11,389,252), filed Jun. 15, 2020 (Jul. 19, 2022), Rotating Marker for Image Guided Surgery. cited by applicant
U.S. Appl. No. 18/008,980, filed Dec. 8, 2022, Rotating Marker. cited by applicant
U.S. Appl. No. 17/368,859 (U.S. Pat. No. 11,896,445), Jul. 7, 2021 (Feb. 13, 2024), Iliac Pin and Adapter. cited by applicant
U.S. Appl. No. 18/437,898, filed Feb. 9, 2024, Iliac Pin and Adapter. cited by applicant
U.S. Appl. No. 18/576,516, filed Jan. 4, 2024, Iliac Pin and Adapter. cited by applicant
U.S. Appl. No. 17/388,064, filed Jul. 29, 2021, Rotating Marker and Adapter for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/291,731, filed Jan. 24, 2024, Rotating Marker and Adapter for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/365,844, filed Aug. 4, 2023, Augmented-Reality Surgical System Using Depth Sensing. cited by applicant
U.S. Appl. No. 18/683,676, filed Feb. 14, 2024, Stereoscopic Display and Digital Loupe for Augmented-Reality Near-Eye Display. cited by applicant
U.S. Appl. No. 18/683,680, filed Feb. 14, 2024, Augmented Reality Assistance for Osteotomy and Discectomy. cited by applicant
U.S. Appl. No. 18/684,756, filed Feb. 19, 2024, Registration and Registration Validation in Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/693,338, filed Mar. 19, 2024, Surgical Planning and Display. cited by applicant
U.S. Appl. No. 18/365,566, filed Aug. 4, 2023, Systems for Medical Image Visualization. cited by applicant
U.S. Appl. No. 18/398,837, filed Dec. 28, 2023, Adjustable Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant
U.S. Appl. No. 18/399,433, filed Dec. 28, 2023, Configurable Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant
U.S. Appl. No. 35/508,942 (D. 930,162), filed Feb. 13, 2020 (Sep. 7, 2021), Medical Headset. cited by applicant
16 Augmented Reality Glasses of 2021 (with Features), in Back to News, Dated May 6, 2022, accessed at <https://web.archive.org/web/20221127195438/https://circuitstream.com/blog/16-augmented-reality-glasses-of-2021-with-features-breakdowns/>. cited by applicant
Eversight, Installing your RX Adaptor, accessed Mar. 13, 2024 at <https://support.eversight.com/hc/en-us/articles/115000984571-Installing-your-RX-Adaptor>. cited by applicant
Eversight, Raptor User Manual, copyright 2017, in 46 pages. cited by applicant
Frames Direct, InSpatialRx Prescription Insert, Prescription Insert for Magic Leap 1, accessed Mar.

8, 2024 at <https://www.framesdirect.com/inspatialrx-prescription-insert.html>. cited by applicant

ISA/220—Notification of Transmittal or Search Report and Written Opinion of the ISA, or the Declaration Mailed on Aug. 11, 2023 for WO Application No. PCT/IB23/054056, 19 page(s). cited by applicant

Reddit, Notice on Prescription Lenses for Nreal Glasses, accessed Mar. 13, 2024 at https://www.reddit.com/r/nreal/comments/x1fte5/notice_on_prescription_lenses_for_nreal_glasses/. cited by applicant

Vuzix Blades, Prescription Lens Installation Guide, copyright 2020. cited by applicant

U.S. Appl. No. 18/857,558, Oct. 17, 2024, Reduction of Jitter in Virtual Presentation. cited by applicant

U.S. Appl. No. 18/398,837, Dec. 28, 2023, Adjustable Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant

U.S. Appl. No. 18/399,433, Dec. 28, 2023, Configurable Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant

U.S. Appl. No. 18/772,578, Jul. 15, 2024, Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant

U.S. Appl. No. 35/508,942, Feb. 13, 2020, Medical Headset. cited by applicant

Augmedics Ltd., 510k Clearance Summary for Augmedics' xvision Spine system, dated Dec. 20, 2019 in 11 pages. cited by applicant

Medtronic Navigation, Inc., StealthStation™ S8 System Manual in 82 pages, Revision 2, Copyright 2018. cited by applicant

Novarad Healthcare IT and Imaging, OpenSight ENGLISH: See 3D Medical Images Using Augmented Reality, dated Mar. 9, 2018, accessed via YouTube on Mar. 11, 2025 at https://www.youtube.com/watch?v=M3yY_b8jT54. cited by applicant

Primary Examiner: Wilson; Nicholas R

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/365,566, filed Aug. 4, 2023, which is a continuation of International PCT Application PCT/IB2023/054056, filed Apr. 20, 2023, which claims the benefit of U.S. Provisional Patent Application No. 63/333,128, filed Apr. 21, 2022 and of U.S. Provisional Patent Application No. 63/428,781, filed Nov. 30, 2022. The entire contents of each of the foregoing applications is incorporated herein by reference.

FIELD

(1) The present disclosure generally relates to medical visualization, including systems and methods for generating (e.g., computing) three-dimensional (3D) models based on 2D or 3D medical images for use, among other considered medical usages, such as in image-guided surgery, and for generating 3D renderings based on the 2D or 3D medical images (e.g., for output on an augmented reality display).

BACKGROUND

(2) Near-eye display devices and systems, such as head-mounted displays are commonly used in augmented reality systems, for example, for performing image-guided surgery. In this way, a computer-generated three-dimensional (3D) model of a volume of interest of a patient may be presented to a healthcare professional who is performing the procedure, such that the 3D model is

visible to the professional engaged in optically viewing an anatomical portion of a patient who is undergoing the procedure.

(3) Systems of this sort for image-guided surgery are described, for example, in Applicant's U.S. Pat. Nos. 9,928,629, 10,835,296, 10,939,977, PCT International Publication WO 2022/053923, and U.S. Patent Application Publication 2020/0163723. The disclosures of all of these patents and publications are incorporated herein by reference.

SUMMARY

(4) In accordance with several embodiments, systems, devices and methods are described that provide enhanced or improved display of 3D models in connection with image-guided medical procedures. For example, the 3D models may be displayed with reduced noise and increased image quality. In some instances, the 3D models may not be displayed with background features (e.g., soft tissue is not displayed when it is bone tissue that is desired to be displayed). In some instances, the 3D models that are generated include implants or hardware (such as screws, rods, cages, pins, tools, instruments, etc.) but not certain types of tissue (e.g., soft tissue, nerve tissue) that is not the focus of the particular medical procedure. For example, the content of the 3D model or rendering that is generated for display may preferentially or selectively include only certain types of content that a clinical professional or other operator would want to see and not “background” content that the clinical professional or other operator does not need to see or does not want to see because it does not impact or affect the medical procedure (e.g., surgical or non-surgical therapeutic procedure or diagnostic procedure). In the example of a spinal surgical procedure, the background content that may be desired to be filtered out or selectively or preferentially not displayed may be soft tissue surrounding or between the vertebrae of the spine and the content that may selectively or preferentially be displayed is the bone tissue (and optionally, any hardware, implant or instruments or tools within the bone, such as screws, rods, cages, etc.).

(5) In accordance with several embodiments, the systems, devices and methods described herein involve automatically segmenting the image(s) (e.g., 3D computed tomography images, 3D magnetic resonance images, other 2D or 3D images) received from an imaging device scan prior to applying one or more 3D model generation algorithms or processes. In accordance with several embodiments, the segmentation advantageously reduces noise around the bone structure and provides a better “starting point” for the 3D model generation algorithms or processes. The segmentation may involve segmenting the 3D image into multiple separate regions, sections, portions, or segments. An entire anatomical portion of a subject (e.g., an entire spine or entire other bone, such as a hip, knee, shoulder, ankle, limb bone, cranium, facial bone, jaw bone, etc.) may be segmented or a subportion of the anatomical portion may be segmented.

(6) Several embodiments are particularly advantageous because they include one, several or all of the following benefits: (i) using a segmented image for 3D image rendering and/or model building to achieve a more accurate and less noisy 3D model and/or visualization of a patient anatomy (e.g., a portion of the patient anatomy); and/or (ii) using artificial intelligence based segmentation in 3D model building to support low-quality intra-operative scanners or imaging devices; and/or (iii) applying different threshold values to different portions of the medical image to achieve better visualization of the patient anatomy; and/or (iv) enhancing a 3D model building while using a model building algorithm which may receive only a single threshold by applying multiple thresholds and/or (v) allowing a user to select different thresholds to be applied on different portions of a medical image (e.g., to improve or optimize visualization).

(7) In accordance with several embodiments, a system for improving display of 3D models in connection with image-guided medical procedures (e.g., surgical or non-surgical therapeutic and/or diagnostic procedures) comprises or consists essentially of a wearable device (e.g., head-mounted unit such as eyewear) including at least one see-through display configured to allow viewing of a region of a body of a patient through at least a portion of the display and at least one processor (e.g., a single processor or multiple processors) configured to perform actions (e.g., upon execution

of stored program instructions on one or more non-transitory computer-readable storage media). For example, the at least one processor is configured to receive a three-dimensional (3D) image of the region of the body of the patient, the 3D image having intensity values; segment the 3D image to define at least one region of interest (ROI) of the region of the body (e.g., a portion of a spine or other bone associated with the medical procedure); determine at least one ROI intensity threshold value of the at least one ROI; determine a background intensity threshold value; and generate a 3D rendering of the 3D image.

(8) The generation of the 3D rendering may include, in the at least one defined ROI of the 3D image, rendering based on intensity values of the at least one ROI that satisfy a lowest threshold value of the at least one ROI intensity threshold value and the background intensity threshold value; and, in a background region of the 3D image, rendering based on intensity values of the background region that satisfy the background intensity threshold value. The background region of the 3D image may include a portion of the 3D image which is not an ROI.

(9) The at least one processor may also be configured to cause the 3D rendering to be output to the display of the wearable device.

(10) In some embodiments, the intensity values are values of 3D image voxels of the 3D image.

(11) In some embodiments, the wearable device is a pair of glasses or other eyewear. In some embodiments, the wearable device is an over-the-head mounted unit, such as a headset.

(12) In some embodiments, the wearable device is configured to facilitate display of 3D stereoscopic images that are projected at a distance to align with a natural focal length of eyes (or a natural convergence or focus) of a wearer of the wearable device to reduce vergence-accommodation conflict.

(13) The 3D image may be a computed tomography image, a magnetic resonance image, or other 3D image generated by another 3D imaging modality.

(14) In some embodiments, the at least one processor is configured to display the 3D rendering in alignment by performing registration of the 3D rendering with the body region of the patient (e.g., using one or more markers, such as retroreflective markers that can be scanned or imaged by an imaging device of the wearable device).

(15) In some embodiments, the 3D rendering is a 3D model. The 3D rendering may be output for display as a virtual augmented reality image. The virtual augmented reality image may be projected directly on a retina of the wearer of the wearable device. The virtual augmented reality image may be presented in such a way that the wearer can still see the physical region of interest through the display.

(16) In some embodiments, the at least one processor is configured to generate the 3D rendering by changing the determined intensity values of the 3D image into a value which does not satisfy the lowest threshold value.

(17) In some embodiments, the background region of the 3D image includes soft tissue.

(18) In some embodiments, the at least one processor is further configured to repeatedly adjust the at least one ROI intensity threshold value and the background feature intensity threshold value according to input from a user; and repeatedly generate a 3D rendering of the 3D image based on the adjusted values of the at least two intensity thresholds. Only one of the threshold values may be adjusted in some embodiments.

(19) In accordance with several embodiments, a system for improving display of 3D models in connection with image-guided surgery comprises or consists essentially of a head-mounted unit including at least one see-through display configured to allow viewing of a region of a spine of a patient through at least a portion of the display and at least one processor configured to (e.g., upon execution of program instructions stored on one or more non-transitory computer-readable storage media): receive a three-dimensional (3D) image of the region of the spine of the patient, the 3D image having intensity values, segment the 3D image to define multiple regions of interest (ROI) of the spine (e.g., multiple vertebrae or multiple vertebral segments); determine a ROI intensity

threshold value for each of the multiple ROIs of the spine; determine a background intensity threshold value; generate a 3D rendering (e.g., 3D model) of the 3D image; and cause the 3D rendering to be output to the display as a virtual augmented reality image. The generation of the 3D rendering includes, in the multiple ROIs of the 3D image, rendering based on the determined ROI intensity values of the multiple ROIs that satisfy a lowest threshold value of the intensity threshold value and the background intensity threshold value and, in a background region of the 3D image, rendering based on intensity values of the background region that satisfy the background intensity threshold value. The background region of the 3D image includes a portion of the 3D image which is not an ROI.

(20) In some embodiments, the intensity values are voxel values of the 3D image.

(21) In some embodiments, the head-mounted unit is a pair of glasses or other form of eyewear, including eyewear without lenses. In some embodiments, the head-mounted unit is an over-the-head mounted unit (e.g. a headset).

(22) In some embodiments, the display is configured to be displayed directly on a retina of a wearer of the head-mounted unit.

(23) In some embodiments, the at least one processor is configured to display the 3D rendering in alignment by performing registration of the 3D rendering with the region of the spine.

(24) In some embodiments, the at least one processor is configured to generate the 3D rendering by changing the determined intensity values of the 3D image into a value which does not satisfy the lowest threshold value.

(25) In some embodiments, the at least one processor is further configured to repeatedly adjust the at least one ROI intensity threshold value and the background feature intensity threshold value according to input from a user and repeatedly generate a 3D rendering of the 3D image based on the adjusted values of the at least two intensity thresholds.

(26) An embodiment of the present disclosure that is described hereinafter provides a computer-implemented method that includes obtaining a three-dimensional (3D) image of a region of a body of a patient, the 3D image having feature values. The 3D image is segmented to define one or more regions of interest (ROIs). The one or more regions of interest may include one or more portions of a bone or joint (e.g., a portion of a spine, individual vertebrae of a portion of a spine, a particular spinal segment, a portion of a pelvis or sacroiliac region, or other bone or joint). At least one region of interest (ROI) feature threshold is determined. A background feature threshold is determined. A 3D model is generated from the 3D image based on the determined at least one ROI feature threshold, the determined background feature threshold, and the segmentation. The 3D model is outputted for display to a user.

(27) In some embodiments, the feature values are intensity values, the ROI feature threshold is an ROI intensity threshold, and the background feature threshold is a background intensity threshold.

(28) In some embodiments, the intensity values are the 3D image voxel values, and the intensity thresholds are the 3D image voxel thresholds.

(29) In an embodiment, the method further includes generating a 3D rendering of the 3D image, wherein the generation of the 3D rendering includes (i) in the at least one ROI of the 3D image, rendering based on feature values of the at least one ROI that satisfy the lowest threshold of the at least one ROI feature threshold and the background feature threshold, and (ii) in a background region of the 3D image, rendering based on feature values of the background region that satisfy the background feature threshold, wherein the background region of the 3D image includes a portion of the 3D image which is not an ROI. The 3D model is outputted for display to a user.

(30) In another embodiment, generating the 3D model includes using a 3D model generation algorithm and providing a feature threshold as input to the 3D model generation algorithm.

(31) In some embodiments, the 3D model generation algorithm is a marching cubes algorithm.

(32) In some embodiments, the provided feature threshold corresponds to one of the at least one ROI feature threshold.

- (33) In some embodiments, the provided feature threshold is determined based on the at least one ROI feature threshold and the background feature threshold.
- (34) In other embodiments, the provided feature threshold is selected as the lowest of the at least one ROI feature threshold value and the background feature threshold value.
- (35) In some embodiments, the generation of the 3D model includes selecting feature values of the 3D image satisfying the provided feature threshold and omitting portions of the background region of the 3D image with selected feature values from being an input to the generation of the 3D model.
- (36) In other embodiments, the omitting of portions of the background region includes changing the selected feature values of the portions of the background region into a value that does not satisfy the provided feature threshold.
- (37) In an embodiment, the determination of the at least one ROI feature threshold and of the background feature threshold includes receiving input values for at least one of: the at least one ROI feature threshold or the background feature threshold.
- (38) In some embodiments, the input values are received for the at least one ROI feature threshold and the background feature threshold.
- (39) In some embodiments, the input values are received from a user.
- (40) In some embodiments, the receiving of the input values from the user includes generating a Graphical User Interface (GUI) element to be displayed to the user, the GUI element allowing the user to adjust the input values, and the rendering and displaying of the 3D rendering is iteratively performed in correspondence to the user adjustment of the input values.
- (41) In an embodiment, in response to a request of the user, the 3D model is generated based on current input values for at least one of: the at least one ROI feature threshold or the background feature threshold.
- (42) In another embodiment, the input values are received for the at least one ROI feature threshold and the background feature threshold.
- (43) In some embodiments, the method further includes displaying a default 3D model based on default values for the at least one ROI feature threshold and the background feature threshold, and displaying the default 3D model to the user.
- (44) In some embodiments, the determining of the at least one ROI feature threshold includes, when multiple ROIs are considered, determining respective multiple ROI feature thresholds.
- (45) In some embodiments, the determining of the at least one ROI feature threshold includes setting a feature threshold value that differentiates bone from soft tissue.
- (46) In other embodiments, the determining of the background feature threshold includes setting a feature threshold value that differentiates metal from bone.
- (47) In an embodiment, the method further includes using the 3D model with an image-guided system. The image-guided system and methods described herein can be surgical, non-surgical or diagnostic.
- (48) In another embodiment, the image-guided system is an augmented or mixed reality system including a direct see-through display, such as a Head Mounted Display (HMD). In some embodiments, the image-guided system is an augmented or mixed reality system that does not include a head mounted display/component or includes both head mounted and non-head mounted displays/components.
- (49) In some embodiments, the segmentation is performed based on one or more deep learning networks. In some embodiments, the deep learning networks are convolutional neural networks.
- (50) There is additionally provided, in accordance with another embodiment of the present disclosure, a computer-implemented method including obtaining a three-dimensional (3D) image of a region of a body of a patient, the 3D image having features. The 3D image is segmented to define one or more regions of interest (ROIs). At least one ROI feature threshold is determined. A background feature threshold is determined. A 3D rendering of the 3D image is generated, wherein

the generation of the 3D rendering includes (i) in the at least one defined ROI of the 3D image, rendering based on feature values of the at least one ROI that satisfy the lowest threshold of the at least one ROI feature threshold and the background feature threshold, and (ii) in a background region of the 3D image, rendering based on feature values of the background region that satisfy the background feature threshold, wherein the background region of the 3D image includes a portion of the 3D image which is not an ROI. The 3D model is outputted for display to a user.

(51) In some embodiments, the method further includes generating a 3D model based on the determined at least one ROI feature threshold, the determined background feature threshold, and the segmentation. The 3D model is displayed on a display.

(52) In some embodiments, generating the 3D model includes using a 3D model generation algorithm and providing a feature threshold as input to the 3D model generation algorithm.

(53) In an embodiment, the provided feature threshold is selected to be the lowest of the at least one ROI feature threshold and the background feature threshold.

(54) In another embodiment, the provided feature threshold is the at least one ROI feature threshold, and the portions of the background region having selected feature values which do not satisfy the background intensity threshold are omitted.

(55) In some embodiments, the determination of the at least ROI feature threshold and of the background feature threshold includes receiving input values for at least one of: the at least one ROI feature threshold or the background feature threshold.

(56) In some embodiments, the method further includes displaying a default 3D rendering based on default values for the at least one ROI feature threshold and the background feature threshold, and displaying the default 3D rendering to the user.

(57) There is further provided, in accordance with another embodiment of the present disclosure, a system for image-guided surgery (or other procedures such as non-surgical procedures and diagnostics), the system including at least one display, and at least one processor. The at least one processor is configured to (i) receive a three-dimensional (3D) image of a region of a body of a patient, the 3D image having intensity values, (ii) compute a 3D model of the region based on at least two intensity threshold values applied to the 3D image, and (iii) cause the computed 3D model to be output for display on the at least one display.

(58) In some embodiments, the at least one display includes a see-through display configured to allow viewing of a body of a patient through at least a portion of the display.

(59) In some embodiments, the see-through display is included in a head-mounted unit. The head-mounted unit may provide elimination of attention shift and a reduced (e.g., minimized) line of sight interruption.

(60) In other embodiments, the computed 3D model is displayed in alignment with the body of the patient as viewed through the see-through display. The 3D model may be provided in a 3D stereoscopic view. The virtual images may be projected at 50 cm in order to align with a normal focal distance of the eyes so as to reduce “vergence-accommodation conflict”, which can result in fatigue, headache, or general discomfort. The line of sight may be at approximately 30 degrees.

(61) In other embodiments, the at least one processor is configured to display in alignment by performing registration of the 3D model with the body of the patient.

(62) In some embodiments, the intensity values are the 3D image voxel values.

(63) In an embodiment, the at least one display includes a stationary display. In some embodiments, the display is a display of a head-mounted device, including but not limited to glasses, goggles, spectacles, visors, monocle, other eyewear, or over-the-head headset. In some embodiments, the head-mounted displays are not used or used together with stand-alone displays, such as monitors, portable devices, tablets, etc. The display may be a hands-free display such that the operator does not need to hold the display.

(64) The head-mounted device may alternatively be a wearable device on a body part other than the head (e.g., a non-head-mounted device). The head-mounted device may be substituted with an

alternative hands-free device that is not worn by the operator, such as a portal, monitor or tablet.

The display may be a head-up display or heads-up display.

(65) In an embodiment, the at least one processor is further configured to segment the 3D image to define one or more regions of interest (ROIs), and the computation of the 3D model is further based on the segmentation.

(66) In another embodiment, the at least two intensity threshold values include at least one ROI threshold intensity value and a background threshold intensity value, and the at least one processor is further configured to (i) compute the 3D model by selecting intensity values of the 3D image satisfying the lowest intensity threshold value of the at least one ROI threshold intensity value, and (ii) omit portions of a background region of the 3D image with intensity values between the lowest intensity threshold value of the at least one ROI threshold intensity values and the background threshold intensity value from being an input to the generation of the 3D model.

(67) In some embodiments, the at least one processor is configured to generate the 3D model by changing the selected intensity values of the 3D image into a value which does not satisfy the lowest intensity threshold value.

(68) In some embodiments, the at least two intensity thresholds include at least one region of interest (ROI) intensity threshold to be applied to one or more ROIs and a background intensity threshold to be applied to at least a background region of the 3D image, and the background region of the 3D image includes a portion of the 3D image which is not an ROI.

(69) In other embodiments, the at least one processor is further configured to (i) repeatedly adjust the at least two intensity thresholds according to input from a user, and (ii) repeatedly generate a 3D rendering of the 3D image based on the adjusted values of the at least two intensity thresholds.

(70) In an embodiment, a method is provided of generating at least one of a 3D model and images for display.

(71) In an embodiment, a system is provided, that includes at least one processor configured for facilitating generation of at least one of a 3D model and images, the system further including a display device configured to display the at least one of a 3D model and images.

(72) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of a spine through a surgical intervention.

(73) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of an orthopedic joint through a surgical intervention, including, optionally, a shoulder, a knee, an ankle, a hip, or other joint.

(74) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of a cranium through a surgical intervention.

(75) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of a jaw through a surgical intervention.

(76) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of a spinal abnormality or degeneration or deformity.

(77) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of a spinal injury.

(78) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of joint damage.

(79) Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of an orthopedic injury.

(80) In accordance with several embodiments, any of the methods described herein may include diagnosing and/or treating a medical condition, the medical condition comprising one or more of the following: back pain, spinal deformity, spinal stenosis, disc herniation, joint inflammation, joint damage, ligament or tendon ruptures or tears.

(81) In accordance with several embodiments, a method of presenting one or more images on a wearable display is described and/or illustrated herein during medical procedures, such as

orthopedic procedures, spinal surgical procedures, joint repair procedures, joint replacement procedures, facial bone repair or reconstruction procedures, ENT procedures, cranial procedures or neurosurgical procedures.

(82) For purposes of summarizing the disclosure, certain aspects, advantages, and novel features of embodiments of the disclosure have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment of the disclosure disclosed herein. Thus, the embodiments disclosed herein may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught or suggested herein without necessarily achieving other advantages as may be taught or suggested herein. The systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein. The methods summarized above and set forth in further detail below describe certain actions taken by a practitioner; however, it should be understood that they can also include the instruction of those actions by another party. Thus, actions such as “adjusting a threshold” include “instructing the adjustment of a threshold.”

(83) The present disclosure will be more fully understood from the following detailed description of the embodiments thereof, taken together with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting features of some embodiments are set forth with particularity in the claims that follow. The following drawings are for illustrative purposes only and show non-limiting embodiments. Features from different figures may be combined in several embodiments. It should be understood that the figures are not necessarily drawn to scale. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated.

(2) FIG. 1 is a schematic, pictorial illustration of an example system for image-guided surgery or other medical intervention utilizing a head-mounted display, in accordance with an embodiment of the disclosure.

(3) FIG. 2 is a flowchart of steps performed to generate a 3D rendering and/or a 3D model based on a 3D medical image of at least a portion of a body of a patient, in accordance with an embodiment of the disclosure.

(4) FIGS. 3A-3C show graphical user interface displays that include 3D renderings based on processed 3D images that facilitate the determination of feature thresholds used for computing respective 3D models, in accordance with embodiments of the disclosure.

(5) FIG. 4 is a graphical user interface display that includes a generated 3D model of a volume of interest of a patient, the 3D model computed based on the 3D image used to generate the 3D rendering of FIG. 3C, in accordance with an embodiment of the disclosure.

(6) FIG. 5 is a flowchart of steps performed to generate a three-dimensional (3D) model based on a 3D image of a portion of a body of a patient using a selected threshold value, in accordance with an embodiment of the disclosure.

(7) FIGS. 6A-6D are example graphical user interface displays that include various views of a patient's spine, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

Overview

(8) A three-dimensional (3D) model of a volume of interest of a patient may include irrelevant or undesired information around the volume of interest or around the model, such as irrelevant or undesired objects or tissue and/or image noise. A user may adjust or cause adjustment of a

threshold (e.g., radiodensity threshold value such as using a Hounsfield Unit scale) of an image (e.g., a CT image or other 2D or 3D medical image), to suppress the irrelevant or undesired information. However, reducing irrelevant or undesired information this way may cause loss of some of the relevant or desired information. For example, a 3D image may include metal objects characterized by very high radiodensity values and bone characterized by lower radiodensity values. Setting a high enough threshold may show the metal objects and remove noise but may well erode or remove some bone structure features in the image.

(9) In accordance with several embodiments, a pre-processing stage may be included to automatically segment the medical image (e.g., 3D CT scan) prior to generation of a 3D model (e.g., generating a 3D model by applying one or more 3D model generation algorithms or processes). The segmentation may advantageously reduce the noise around the bone structure and provide a better “starting point” for the 3D model generation. The segmentation may be applied to the entire medical image or to one or more portions, regions or segments of the medical image.

(10) Image-guided surgery employs tracked surgical instruments and images of the patient anatomy in order to guide the procedure. In such procedures, a proper visualization of regions of interest of the patient anatomy, including tissue of interest, is of high importance. Enhanced or improved visualization may also advantageously provide increased user adoption and enhance user experience.

(11) In particular, during image-guided surgery or other medical intervention (e.g., procedures employing augmented reality technology, virtual reality technology or mixed-reality technology), the 3D model of a volume of interest of the patient upon whom the surgery or other medical intervention (including diagnostic and/or therapeutic intervention) is being performed may be aligned with tools or devices used during the surgery or intervention and/or with the anatomical portion of the patient undergoing surgery or other medical intervention. Such 3D model may be derived from a 3D medical image, such as a 3D CT image or 3D MR image. In accordance with several embodiments, a proper modeling based on the 3D medical image is important for providing the user with the required and accurate information for navigating a medical tool on anatomical images derived from the 3D medical image (e.g., slices, 3D model and/or 2D images) and for accurate and/or proper augmentation (e.g., in alignment) of the 3D model with the optically viewed (e.g., with a near-eye display device) anatomical portion of the patient when augmented-reality is used.

(12) Some embodiments of the present disclosure provide computer-implemented methods for generating a 3D model from at least one 3D medical image. The methods include obtaining one or more 3D images of at least a region of a body of a patient (e.g., including a cervical spine region, a thoracic spine region, a lumbosacral spine region, a cranial region, an ENT-associated region, a facial region, a hip region, a shoulder region, a knee region, an ankle region, a joint region, or a limb region), the 3D image having one or more feature values, such as image intensity values on a gray-scale, RGB values, gradients, edges and/or texture. In some embodiments, the 3D image is first segmented to include or identify one or more regions of interest (ROIs). At least one ROI feature threshold may be determined. At least one background feature threshold may also be determined. In some embodiments, a 3D model is generated from the 3D image, based on the determined at least one ROI feature threshold, the determined at least one background feature threshold, and the segmentation. The 3D model may then be saved for use in image-guided surgery and/or systems, e.g., to be displayed to a user on a display of a wearable (e.g., head-mounted or near-eye or direct see-through) display device and/or on a separate or stationary display monitor or device (e.g., a standalone or separate portal, tablet or monitor).

(13) In some embodiments, the 3D image is processed to only show the at least one ROI and to only show background information above the background feature threshold. This may be performed, for example, by segmenting the image and applying different feature thresholds to different portions of the image. In such embodiments, irrelevant or undesired background

information can be suppressed using a high background feature threshold, without desired features inside the at least one ROI being eroded.

(14) As noted above, the techniques disclosed herein may advantageously present (e.g., cause to be displayed) maximum or an increased amount of information obtained from ROIs of the 3D medical image and filter out some of the background of the 3D medical image, which may include, for example, noise or information of less interest. Accordingly, in some embodiments, the at least one processor may be configured to apply the lowest feature threshold to the ROIs (e.g., even if a specific ROI's feature threshold is not or different than the lowest feature threshold). In some embodiments, the at least one processor may be configured to apply the background feature threshold to the entire image while determining in each region or segment of the image the lowest feature threshold applied to the region or segment as the specific region or segment feature threshold. For example, for a specific ROI, a background feature threshold and an ROI feature threshold are applied. One may expect that the ROI feature threshold would be the lowest but in case the background feature threshold is the lowest, then the background feature threshold value would determine the threshold value for the specific ROI. Specifically, a background feature (e.g., intensity) threshold may be set to a higher value than any of one or more ROI feature (e.g., intensity) thresholds, for example. As a result, the processed 3D image (e.g., processed via volume rendering) may show only information contained in the at least one ROI and background information (e.g., objects) of sufficient intensity, while removing irrelevant or undesired background information. The processed 3D image may show in an ROI bone structure with metal surgical objects in the background, without showing irrelevant anatomy (e.g., other bones or other type of tissue) or an image suffering from background image noise.

(15) In some embodiments, at least two feature thresholds are provided that include at least one region of interest (ROI) feature threshold to be applied to the one or more ROIs, and a background feature threshold to be applied to a background region of the 3D image. The background region of the 3D image may include a portion of the 3D image which is not an ROI or the portion or all portions of the 3D image which are not an ROI. By suppressing the values of the background regions (e.g., values of 3D image voxels) which do not satisfy the background feature threshold, an effective unique threshold may be obtained. This suppression step may be used for generating a 3D model using an algorithm that accepts only one threshold value (e.g., marching cubes algorithm), while at the same time maximizing or otherwise increasing information obtained from ROIs of the 3D medical image and filtering out some of the background of the 3D medical image, which may be an area of less interest.

(16) In some embodiments of the computer-implemented method, before the 3D model is computed, a 3D rendering based on a processed 3D image (e.g., a segmented image) is displayed to a user, such as a surgeon or other healthcare professional. The user may view the 3D rendering on a monitor or a display (e.g., a graphical user interface display on a monitor or on a near-eye display of a head-mounted, direct see-through, or near-eye display device). The graphical user interface display may allow the user to adjust the at least one ROI intensity threshold and/or the background intensity threshold, to thereby modify the look of the 3D rendering. This way, the user can advantageously optimize the rendered 3D image by selecting and/or identifying optimal and/or desired feature thresholds, based on which the 3D model will be computed.

(17) The 3D rendering may be made of the original image voxels, or may be a result of further image processing steps applied to the 3D image, such as surface ray casting. Further image processing of the 3D image beyond producing a 3D rendering using the native 3D image voxels can have advantages. For example, using ray casting may facilitate interpolation between, e.g., voxel intensity values, to produce a smoother look. Also, some methods such as surface ray cast rendering use only a portion of the information included in the native 3D image (e.g., use only voxels encountered by rays, provided the voxels have feature values such as intensities above a threshold). In accordance with several embodiments, using only a subset of the voxels minimizes

or otherwise advantageously reduces computation effort and time required to generate a 3D rendering. This reduction may be particularly important as the user manipulates the view of the rendering (e.g., views the rendering from different points of view), which requires repeated computations of the 3D rendering.

(18) In accordance with several embodiments, the 3D rendering generation process comprises the steps of: In the at least one ROI of the 3D image, rendering based on feature values of the at least one ROI which satisfy the lowest threshold of the at least one ROI feature threshold and the background feature threshold. In a background region of the 3D image, rendering based on feature values of the background region which satisfy the background feature threshold, wherein the background region of the 3D image comprises a portion of the 3D image which is not an ROI.

(19) As noted above, to generate the 3D model, two thresholds (e.g., the ROI feature threshold and the background feature threshold) may be determined by a user (or automatically by at least one processor, e.g., based on accumulated data or based on trained machine learning algorithms or neural networks). However, some 3D model generation algorithms (e.g., surface building algorithms, such as the Marching Cubes algorithm), may receive as input only a single threshold value. In accordance with several embodiments, the techniques disclosed herein advantageously solve the limitation of only a single allowed input threshold by: Using the lowest ROI feature threshold value with the algorithm (e.g., running the Marching Cubes algorithm) to delineate ROI regions. Visualizing only ROI regions with feature values higher than a lowest ROI feature threshold. Visualizing only portions of any other regions, including background regions, with feature values higher than a background feature threshold.

(20) In accordance with several embodiments, background information (e.g., voxel values) having feature values higher than the ROI feature threshold but lower than the background feature threshold is removed (e.g., in order not to model and display irrelevant or undesired information). In some embodiments, such removal is performed by determining the value of such voxels to a value which does not satisfy the ROI feature threshold(s), e.g., equal to or lower than the ROI feature threshold (e.g., ROI threshold-C) while C is a constant (non-negative number): $C > 0$. In some embodiments, the subtracting of such a constant may allow further smoothing of the visualization (e.g., the 3D model).

(21) As described above, using two thresholds, called herein TH ROI and TH Background, may allow showing ROI information and some selected background objects, respectively. If ROI information to be visualized has to meet multiple thresholding conditions (e.g., multiple tissue types have to be visualized, such as fat, muscle and bone), then two or more (e.g., multiple) ROI thresholds (e.g., one for each ROI type), $TH(j) \text{ ROI}$, $j=1, 2, \dots$, may be defined and used in segmentation, volume rendering and/or generating a 3D model. In some embodiments when only one threshold may be input to a 3D model generation algorithm, then the lowest of $\{TH(j) \text{ ROI}\}$ may be used.

(22) In one embodiment, a trained neural-network (NN) based segmentation algorithm is provided. The disclosed segmentation algorithm classifies different regions of 3D image as ROI and background and delineates one from another. For example, in orthopedic image-guided applications (such as surgery or other procedures, such as non-surgical or diagnostic procedures), the disclosed segmentation algorithm may be trained to classify vertebrae column and the Iliac and Sacrum as "Spine". According to some embodiments, the vertebrae, each vertebra, the Iliac and/or the Sacrum may be segmented separately. According to some embodiments, screws, implants and/or inter-bodies included in the scan may also be classified by the algorithm as "Spine" (or the desired category to be shown). Soft tissues, the rib cage and instruments such as clamps, retractors and rods that are included in the scan may be classified as "background." In some implementations, the segmentation of the spine is done on the whole spine anatomy included in the scan. In some implementations, the segmentation of the spine is performed on portions of the spine anatomy (e.g., just the lumbosacral region, just the lumbar region, just the thoracic region, just the sacral region,

just the sacroiliac region, just the cervical region, or combinations thereof) or on individual vertebrae, and/or on anatomical portions of individual vertebrae. The segmentation may further be used for creating a virtual 3D model of the scanned spine area, leaving out, for example, soft tissue, ribs and rods. However, in some embodiments, such left-out elements may be segmented and classified as ROI, such as ribs.

(23) In some embodiments, a segmentation method is applied for generating a 3D model without resorting to a subsequent model building step and algorithm. To this end, the segmentation may use ROI and background feature threshold values in the delineation process. In some embodiments, the segmentation algorithm uses the minimal ROI feature threshold value to segment a first group of regions (e.g., bone regions) in the 3D image, with values above the minimal ROI feature threshold value. Then, the segmentation algorithm uses the background intensity threshold to further segment a second group of regions in the 3D image, with values above the background feature threshold value. A processed image is then generated that visualizes only first and second regions obtained by the segmentation algorithm. If the spatial resolution of the segmentation algorithm is sufficiently high, as well as that of the native 3D image, the segmentation algorithm may be sufficient to create a 3D model (e.g., without resorting to surface building algorithms, such as the Marching Cubes algorithm).

(24) In some embodiments, the at least one processor is configured to use the segmentation to generate a digitally reconstructed radiograph (DRR) of one or more ROIs only. To this end, the at least one processor may be configured to omit the background region, for example, by setting voxel feature values in the medical 3D image (e.g., an input 3D Digital Imaging and Communications in Medicine (DICOM) image) segmented as background to zero. The DRRs may be generated by summarizing feature values in one dimension or by using methods such as Siddon's algorithm published in 1985 by Robert L. Siddon. For example, in Siddon's algorithm, the index of each voxel along a certain projection ray and the interesting length of that ray within that voxel may be computed by four multiplications. Refinements or modifications to Siddon's algorithm, or alternative algorithms (e.g., for calculating radiological paths through pixel or voxel spaces), may also be used.

(25) A further embodiment provides an augmented-reality system for image-guided surgery (or other procedures such as non-surgical procedures and diagnostics), the system comprising (i) a see-through augmented-reality display configured to allow viewing of a body of a patient through at least a portion of the display by the user (e.g., wearing a head-mounted unit), and at least one processor, which is configured to receive the 3D image of a region of the body of the patient and compute a 3D model of the region based on at least two intensity thresholds applied to the 3D image, and present the computed 3D model on the display in alignment with the body of the patient viewed through the display, e.g., for better spatial awareness.

(26) In some embodiments, accurate and/or proper augmentation (e.g., alignment) of a navigated medical tool with images derived from the medical image (e.g., slices, 3D model, 2D images and/or DRR images) and, optionally, accurate and/or proper augmentation of the 3D model with an optically viewed anatomical portion is facilitated by a registration step of the 3D medical image, e.g., via artificial fiducials in the 3D medical image and by utilizing a tracking system. Specifically, when using a direct see-through or head-mounted display, accurate and/or proper augmentation of the 3D model with an optically viewed anatomical portion as viewed from the point of view of a professional wearing the head-mounted display, may be achieved by utilizing a tracking device for tracking at least the position of the head-mounted display relative to the anatomical portion. In some embodiments the tracking system may be mounted on the head-mounted display device. Systems of this sort using augmented-reality for image-guided surgery (or other procedures such as non-surgical procedures and diagnostics), are described, for example, in the above incorporated Applicant's U.S. Pat. Nos. 9,928,629, 10,939,977, PCT International Publication WO 2022/053923, and U.S. Patent Application Publication 2020/0163723.

(27) The disclosure describes, for example, a method of generating a 3D model and/or images for display (e.g., to facilitate image-guided surgery using a direct see-through or head-mounted augmented reality display device) such as described and/or illustrated hereinafter. The disclosure further describes a system comprising at least one processor for generating a 3D model and/or images for display, and a display device configured to display the 3D model and/or images (e.g., to facilitate image-guided surgery using a head-mounted augmented reality display device) such as described and/or illustrated hereinafter.

(28) Finally, as noted above, one common type of 3D medical image in use is a 3D CT image. Another type of 3D medical image that can be used is a 3D MM image. The method can use, mutatis mutandis, 2D or 3D medical images derived from other imaging modalities, such as ultrasound, positron emission tomography (PET), and optical coherence tomography (OCT). Therefore, the above-described methods may be applied to other types of medical acquisitions (e.g., medical scans or images), including electroencephalogram (EEG) imaging, and to other areas or organs of the body, such as skull or cranium, knees, hips, shoulders, sacroiliac joints, ankles, ear, nose, throat, facial regions, elbows, other joint regions or orthopedic regions, gastrointestinal regions, etc. Accordingly, the feature values may be selected as any relevant physiological or electrophysiological values, such as blood and neural activation velocities, respectively, or speeds as derived from the image features.

(29) The systems and methods described herein may be used in connection with surgical procedures, such as spinal surgery, joint surgery (e.g., shoulder, knee, hip, ankle, other joints), orthopedic surgery, heart surgery, bariatric surgery, facial bone surgery, dental surgery, cranial surgery, or neurosurgery. The surgical procedures may be performed during open surgery or minimally-invasive surgery (e.g., surgery during which small incisions are made that are self-sealing or sealed with surgical adhesive or minor suturing or stitching. However, the systems and methods described may be used in connection with other medical procedures (including therapeutic and diagnostic procedures) and with other instruments and devices or other non-medical display environments. The methods described herein further include the performance of the medical procedures (including but not limited to performing a surgical intervention such as treating a spine, shoulder, hip, knee, ankle, other joint, jaw, cranium, etc.).

System Description

(30) FIG. 1 is a schematic, pictorial illustration of an example system **10** for image-guided surgery or other medical intervention utilizing a head-mounted display such as a direct see-through display, in accordance with an embodiment of the disclosure. FIG. 1 is a pictorial illustration of the system as a whole, including showing, by way of example, a head-mounted display unit **28** in a form of eye-glasses, spectacles, or other eyewear that is used in system **10**. Alternative types of a head-mounted unit, such as ones based on a helmet, visor, or over-the-head mounted unit that the surgeon wears, may be used in system **10**. Alternatively, other wearable or hands-free devices, units or displays may be used.

(31) In accordance with several embodiments, system **10** is applied in a medical procedure on a patient **20** using image-guided intervention. In this procedure, a tool **22** is inserted via an incision (e.g., minimally-invasive or self-sealing incision) in the patient's back in order to perform a surgical intervention. Alternatively, system **10** and the techniques described herein may be used, mutatis mutandis, in other surgical or non-surgical medical or diagnostic procedures.

(32) In the pictured embodiment, a user of system **10**, such as a healthcare professional **26** (e.g., a surgeon performing the procedure), wears the head-mounted display unit **28**. In various embodiments, the head-mounted display unit **28** includes one or more see-through displays **30**, for example as described in the above-mentioned U.S. Pat. No. 9,928,629 or in the other patents and applications cited above that are incorporated by reference. Such see-through displays may include an optical combiner that is controlled by a computer processor (e.g., a computer processor **32** in a central processing system **50** and/or a dedicated computer processor in the head-mounted display

unit **28**) to display an augmented-reality image to the healthcare professional **26**. The image is presented on a see-through display **30** (e.g., as a stereoscopic near-eye display) such that a computer-generated image is projected in alignment with the anatomy of the body of patient **20** that is visible to professional **26** through a portion of the see-through display. In some embodiments, one or more projected computer-generated images include a virtual image of tool **22** overlaid on a virtual image of at least a portion of the patient's anatomy (such as slices, the 3D model or 2D images derived from the 3D medical image, e.g., of a portion of the spine). Specifically, a portion of the tool **22** that would not otherwise be visible to the healthcare professional **26** (for example, by virtue of being hidden by the patient's anatomy) is included in the computer-generated image.

(33) In image-guided surgery or in surgeries utilizing augmented reality systems, a bone-anchoring device may be used as a fiducial marker or may be coupled with such a marker, for indicating the patient's body location in a coordinate system. In system **10**, for example, an anchoring device is coupled with a marker that is used to register a region of interest (ROI) of the patient body with a CT scan or other medical images of the ROI (e.g., preoperative, or intraoperative CT or fluoroscopic image). During the procedure, a tracking system (e.g., an IR tracking system of the head-mounted display unit **28**) tracks a marker mounted on the anchoring device and a tool that includes a tool marker. Following that, the display of the CT or other medical image data (which may include, for example, a model generated based on such data) on a near-eye display may be aligned with the professional's actual view of the ROI based on this registration. In addition, a virtual image of the tool **22** may be displayed on the CT or other 3D image model based on the tracking data and the registration. The user (e.g., professional **26**) may then navigate the tool **22** based on the virtual display of the tool **22** with respect to the patient image data, and optionally, while the 3D model is aligned with the professional's view of the patient or ROI.

(34) According to some aspects, the 3D image or model presented on see-through display **30** is aligned with the patient's body. According to some aspects, allowed alignment error (or allowed misalignment) may not be more than about two to three mm or may be less than four mm or less than five mm or about one to two mm or less than one mm. In one embodiment, the allowed alignment error is less than one mm. In order to account for such a limit on error in alignment of the patient's anatomy with the presented images, the position of the patient's body, or a portion thereof, with respect to the head-mounted unit may be tracked. According to some aspects, the 3D model presented on see-through display **30** is misaligned with the patient's body (e.g., having a misalignment greater than the allowed misalignment).

(35) A patient marker **60** attached to an anchoring implement or device such as a clamp **58** or a pin, for example, may be used for this purpose, as described further hereinbelow. In some embodiments, a registration marker, such as registration marker **38**, may be used for registering the 3D image with the patient anatomy, e.g., in a preceding registration procedure. In some embodiments registration marker **38** may be removed once the registration is complete. According to some embodiments, patient marker **60** and registration marker **38** may be combined into a single marker. Anchoring devices, markers and registration systems and methods of this sort for image-guided surgery are described, for example, in the above incorporated U.S. Pat. Nos. 9,928,629, 10,835,296 and 10,939,977, and in addition in Applicant's U.S. Patent Application Publication 2021/0161614, U.S. Patent Application Publication 2022/0142730, U.S. Patent Application Publication 2021/0386482, U.S. Patent Application Publication 2023/0009793, and PCT International Publication WO 2023/026229, all of which are incorporated herein by reference.

(36) When an image of tool **22** is incorporated into the image that is displayed on head-mounted display unit **28**, the position of the tool **22** with respect to the patient's anatomy should be accurately reflected. For this purpose, the position of the tool **22** or a portion thereof (e.g., a tool marker **40**) is tracked by system **10** (e.g., via a tracking system built into the head-mounted display unit **28**). In some embodiments, it is desirable to determine the location of the tool **22** with respect to the patient's body such that errors in the determined location of the tool **22** with respect to the

patient's body are typically less than three mm or less than two and a half mm or less than two mm. (37) In some embodiments, head-mounted display unit **28** includes a tracking sensor **34** to facilitate determination of the location and orientation of head-mounted display unit **28** with respect to the patient's body (e.g., via patient marker **60**) and/or with respect to tool **22** (e.g., via a tool marker **40**). Tracking sensor **34** can also be used in finding the position and orientation of tool **22** (e.g., via tool marker **40**) with respect to the patient's body (e.g., via patient marker **60**). In one embodiment, tracking sensor **34** comprises one or more image-capturing or image-acquiring devices, such as a camera (e.g., infrared camera, visible light camera, and/or an RGB-IR camera), which captures or acquires images of patient marker **60** and/or tool marker **40**, and/or other landmarks or markers. The tracking sensor **34** may comprise an optical tracking device, an RFID reader, an NFC reader, a Bluetooth sensor, an electromagnetic tracking device, an ultrasound tracking device, or other tracking device.

(38) In the pictured embodiment, system **10** also includes a tomographic imaging device, such as an intraoperative computerized tomography (CT) scanner **41**. Alternatively or additionally, processing system **50** may access or otherwise receive tomographic data from other sources; and the CT scanner **41** itself is not necessarily an essential part of the present system **10** (e.g., it is optional). Regardless of the source of the tomographic data, processor **32** computes a transformation over the ROI so as to register the tomographic images with images and information (e.g., a depth map) that are displayed on head-mounted display unit **28**. The processor **32** can then apply this transformation in presenting at least a part of the tomographic image on see-through display **30** in registration with the ROI viewed through the see-through display **30**. In the disclosed technique, the processor **32** can generate from the tomographic images, collectively named 3D image, a 3D model, and register the 3D model with the ROI, e.g., viewed through the see-through display **30**.

(39) In accordance with several implementations, in order to generate and present an augmented reality image on see-through display **30**, processor **32** computes the location and orientation of head-mounted display unit **28** with respect to a portion of the body of patient **20** (e.g., the patient's back or a portion thereof). In some embodiments, processor **32** also computes the location and orientation of tool **22** with respect to the patient's body. In one embodiment, a processor which is integrated within the head-mounted display unit **28**, may perform these functions or a portion of these functions. Alternatively or additionally, processor **32**, which is disposed externally to the head-mounted display unit **28** and which may be in wireless communication with the head-mounted display unit **28**, may be used to perform these functions or a portion of these functions. Processor **32** can be part of processing system **50**, which additionally includes an output device **52** (e.g., a display, such as a monitor) for outputting information to an operator of the system, memory, and/or an input device **54** (such as a pointing device, a keyboard, a mouse, a trackpad, a pedal or touchscreen display, etc.) to allow the operator (e.g., professional **26** or an assistant to professional **26**) to input data into the system **10**.

(40) In general, in the context of the present description, when a computer processor is described as performing certain steps, these steps may be performed by external computer processor **32** and/or a computer processor that is integrated within the head-mounted display unit **28**. The processors described herein (e.g., processor **32**) may include a single processor or multiple processors (e.g., parallel processors running in parallel to reduce computing time). If multiple processors are used, they may be communicatively coupled to each other over a network (e.g., wired or wireless communications network). The processor or processors carry out the described functionality under the control of suitable software, which may be downloaded to system **10** in electronic form, for example over a network, and/or stored on tangible, non-transitory computer-readable media, such as electronic, magnetic, or optical memory. In some embodiments, processing system **50** may be integrated in head-mounted display unit **28**. In some embodiments, system **10** may include a see-through and/or augmented reality display which is not head-mounted. In such systems the display

may be mounted to patient **20** or to the operating table and positioned such that professional **26** may be able to view patient **20** anatomy and/or operation site through at least a portion of the display. The display position may be adjustable.

(41) In some embodiments, the 3D image, 3D rendering and 3D model of the present disclosure may be displayed while allowing user interaction as described herein on a display which is not a see-through display and/or a head-mounted display, such as a display of a workstation, a personal computer, a terminal, a tablet or a mobile device.

Visualizing a 3D Medical Image

(42) FIG. 2 is a flowchart of steps performed to visualize a 3D medical image of a portion of a body of a patient, in accordance with an embodiment of the disclosure. The process can run automatically without user input, by one or more processors running all of the steps. Optionally, a user, such as the professional **26**, may give input in thresholds obtaining step **206** and/or optical image cropping step **218**, as described below. In some embodiments, the processor can be more than one processor, such that various steps are performed by different processors or processing units. In some embodiments, a single processor runs all of the steps. Thus, when a processor is mentioned herein, it may be referring to a single processor or multiple processors or processing units.

(43) The process begins by the processor obtaining a 3D medical image (e.g., a CT scan) of a volume of the body portion, at a medical image obtaining step **202**.

(44) In general, e.g., depending on the clinical application, various medical image formats or modalities may be considered, such as MRI, PET or ultrasound images. In the contexts of this disclosure, the term “obtaining” may refer to, inter alia, acquiring, receiving, and/or accessing the 3D image and does not necessarily require the system to include the imaging device and/or the method to include the image acquiring operation or step.

(45) Next, the processor segments the medical image to delineate at least one ROI (e.g., a spine section or region) from the rest of the imaged volume, at an image segmentation step **204**. Various segmentation methods may be used, such as one that uses a trained convolutional neural network (CNN, such as U-net or V-net).

(46) In some embodiments, a 3D image of the spine is segmented into spinal regions (e.g., cervical, thoracic, lumbar, sacral). In some embodiments, a 3D image of the spine is segmented into individual 3D vertebrae. In some embodiments, a 3D image of the spine is further segmented to indicate anatomical regions or portions of a vertebra (for example, spinous process, articular processes, transverse processes, vertebral body, centrum, posterior vertebral arch or neural arch). Bones other than the bones of the spine or other tissue may also be segmented into various regions, into individual components, and/or into anatomical portions of individual components. Such segmentation, although requiring more computation resources, may provide a better segmentation of the spine or any other such complex structure. This segmentation operation can advantageously be carried out by deep learning techniques, using one or more trained CNNs. The sacrum and ilium may be segmented in this manner, as well, using a separate neural network from the neural network used for the vertebrae or the same neural network.

(47) In some embodiments, a combination of three networks may be used for this purpose. The networks are fully convolutional networks, e.g., based on the U-Net architecture.

(48) The processor may obtain at least one ROI threshold and background threshold levels or values, TH1 ROI and TH2 background, respectively, at thresholds obtaining step **206**. Examples of different ROI intensity threshold and background intensity threshold levels used are described in connection with FIGS. 3A-3C below. In the examples given by FIGS. 3A-3C, the ROI and background thresholds are HU thresholds that govern CT 3D visualization of mainly bone and metal. In these figures, of an orthopedic application concerning a spine, the main aim of the background threshold is to visualize metal elements that are in the background section (e.g., external to the spine) such as clamps and retractors without adding irrelevant or undesired

information (e.g., background noise, such as noisy voxel levels, having a radiodensity value which does not satisfy the background threshold).

(49) In another example, relevant to another medical application, the ROI and background thresholds can be MM intensities set to optimize, for example, an MRI 3D visualization of a joint or of at least a portion of a brain. In other embodiments, to, for example, distinguish between multiple tissue types, two or more (e.g., multiple) ROI thresholds can be defined and used together with the background threshold in segmentation, rendering a 3D image and/or generating a 3D model.

(50) In an image processing step **210**, the processor utilizes the segmentation and/or the thresholds to generate a processed 3D image. The processing may include reducing the feature values of voxels classified as background voxels having feature values which satisfy the ROI feature threshold value but do not satisfy the background feature threshold value. Thus, a 3D model generated based on the processed image (e.g., image with altered feature values) and the ROI feature threshold may visualize tissue features inside ROIs (e.g., bone tissue) when irrelevant or undesired background information is suppressed.

(51) The processed 3D image of image processing step **210** may be used by the processor for computing a 3D model, at a 3D model creation step **212**. At 3D model creation step **212**, the processor, in some embodiments, runs a 3D model generation application (e.g., a surface building algorithm), such as a marching cubes algorithm described by Lorensen E. and Harvey E. in a paper titled, "Marching cubes: A high resolution 3D surface construction algorithm," published in ACM SIGGRAPH Computer Graphics, Volume 21 (4): 163-169, July 1987, the content of which is incorporated herein by reference. As described above, the marching cubes algorithm may use a threshold value, e.g., the lowest TH ROI value, to generate the 3D model. In some embodiments, the 3D model comprises a surface of an anatomy in the ROI (e.g., spine) and possibly comprising artificial elements as well, depending on TH ROI and TH background threshold settings. An example of a generated 3D model is shown in FIGS. 3A-3C, described below.

(52) 3D model cropping step **222** and 3D model saving step **224** include generating a suitable file of the 3D model, to be used, for example, with a near eye display (e.g., of head-mounted display unit **28**) in image-guided surgery and/or with any other display such as a work station display. In an optional 3D model cropping step **222**, the processor may define or may receive a definition (e.g., via user input) of a cropping box, such as shown in FIG. 4, and may crop the 3D model accordingly. Finally, the processor saves the 3D model (e.g., cropped 3D model) into a file of a suitable format (e.g., PLY format), in a memory, to be later uploaded to the processor or any other processor (e.g., a dedicated processor of a near-eye display of a head-mounted display unit **28**).

(53) In optional 3D image rendering step **208**, the 3D image obtained medical image obtaining in step **202** is displayed on a monitor as a 3D visualization (e.g., a 3D volume rendering). A user, such as a medical professional can optimize the 3D visualization for the required or desired medical application (e.g., for spine image-guided surgery) by adjusting the two feature thresholds obtained (e.g., received, uploaded, etc.) in thresholds obtaining step **206**. According to some embodiments, the initial 3D rendering is performed based on predetermined default values for the TH1 ROI and TH2 Background. According to some embodiments, the thresholds may be iteratively adjusted by the user. Final threshold values may be set or determined, for example, once a user input indicating that the threshold adjustment process is complete or a user request for generating of the 3D model is received.

(54) In optional image cropping step **218**, the processed 3D image obtained in image processing step **210** is displayed on a monitor so that the user can crop the portion of processed 3D image inputted for the modeling step (e.g., instead of the processor performing this step as part of, for example, image processing step **210**). According to some embodiments, the cropping may be performed following optional 3D image rendering step **208** or in the frame of optional 3D image rendering step **208** and/or prior to image processing step **210**.

(55) In an additional optional step (not illustrated) the generated 3D model is displayed on a display (e.g., a stationary display such as a workstation and/or a near-eye display of head-mounted display unit **28**) for the user's review.

(56) The flowchart of FIG. **2** is presented by way of example and is simplified for clarity of presentation. The process may include additional or alternative steps, such as generating a 3D model from another imaging modality (e.g., based on a volume rendering obtained using ultrasound or MRI or fluoroscopy), or directly from high resolution segmentation, skipping the step of volume rendering, as described above.

(57) FIGS. **3A-3C** are exemplary Graphical User Interface (GUI) displays showing 3D renderings based on processed CT images (e.g., segmented CT images) of a spine that can be used for selecting or determining optimal or desired ROI intensity threshold and background intensity threshold for computing the 3D model, in accordance with embodiments of the disclosure. The 3D renderings may be viewed by a user that can determine or adjust the two thresholds of the thresholds obtaining step **206** above. The user may set or adjust the ROI and background intensity thresholds by adjusting a GUI element, such as sliding ROI TH sliders (**302**, **312**, **322**) and background TH sliders (**304**, **314**, **324**), respectively, at the bottom of the 3D rendering screen, respectively. The 3D rendering will change in correspondence to the change in thresholds and based on the segmentation, visualizing the data that will be included in the 3D model for each selection of thresholds or for each selected combination of thresholds. The user may then optimize the visualization or rendering while the thresholds values for the optimized or desired rendering will be provided as input to the 3D image processing and/or modeling. In FIGS. **3A-3C**, the value of a threshold increases as the threshold slider is slid to the left. The slider may be moved by touching and dragging on the screen and/or by use of a computer mouse, keyboard, joystick, voice-activated control, or other input device.

(58) FIG. **3A** is a graphic display of a rendering **310** of a 3D image with threshold sliders set as follows: ROI TH slider **302** at 100%, meaning ROI intensity TH value=min scan radiodensity value and all is shown within the segmented area, and background threshold slider **304** at 0%, meaning background intensity TH value=maximum scan radiodensity value and all information outside the segmented portion is blocked.

(59) FIG. **3B** is a graphic display of a rendering of a 3D image with threshold sliders set as follows: ROI threshold slider **312** is set at 100%, meaning ROI intensity TH value=minimum scan radiodensity value and all tissue is shown within segmented portion. Background threshold slider is set at 22%, meaning background intensity TH value is high but such that metal retractors **318** and clamp **320** outside the segmented spine are shown.

(60) FIG. **3C** is a graphic display of a rendering of a 3D image with threshold sliders set as follows: ROI threshold slider is set at 92%, meaning ROI intensity TH value=low scan radiodensity value and most tissue is shown within segmented portion. Background threshold slider is set at 47%, meaning background intensity TH value is of mid value but such that the heads of metal screws **340** implanted in the bone protruding from the spine are shown.

(61) Due to the relatively large background threshold, e.g., that is set at about the middle of the intensity range, irrelevant or undesired signals in the background, such as image noise and soft and bone tissue are not shown, making the disclosed processed 3D image particularly well visualizing the ROIs only.

(62) In some embodiments, in generating the rendering, the one or more ROI feature thresholds would be applied to the one or more ROIs respectively while the background feature threshold would be applied only to the background portion of the image.

(63) In some embodiments, as illustrated in FIGS. **3A-3C**, when generating the rendering, the one or more ROI feature thresholds would be applied to the one or more ROIs respectively, while the background feature threshold would be applied to the entire image (i.e., including the one or more ROIs). In such a case, in the one or more ROIs, rendering may be performed based on the lowest

threshold applied (i.e., the lowest of the respective ROI feature threshold and the background feature threshold). In such embodiment, the user may view a rendering or visualization of the medical image without the effect of the segmentation by simply setting the background feature threshold to a value lower than or equal to the one or more ROI feature thresholds. Following that, a single feature threshold value would apply to the entire scan or image without a distinction between different portions of the image (e.g., ROI vs. background).

(64) FIG. 4 is a graphical user interface display showing a 3D model **400** of a volume of interest, the 3D model **400** computed based on processed 3D image of FIG. 3C, in accordance with an embodiment of the disclosure. As seen, the 3D model **400** is located inside a box **405** having marked facets **415**. The disclosed 3D model clearly presents a spine section implanted with metal screws **440** with clean background (e.g., empty of, or without, irrelevant or undesired information). Such 3D model is suitable for presentation and/or augmentation with the optically viewed (e.g., with the near-eye display device) anatomical portion of the patient.

(65) FIG. 5 is a flowchart of steps performed to generate a three-dimensional (3D) model based on a 3D image of a portion of a body of a patient using selected threshold values, in accordance with an embodiment of the disclosure. The method according to the presented embodiment carries out a process that begins with obtaining a 3D medical image having feature values, of at least a portion of an organ or anatomy of a patient (e.g., an entire spine or region of a spine, such as a lumbosacral region, a cervical region, a thoracic region, and/or a sacroiliac region), at a 3D medical image receiving step **502**. The features may be directly extracted from the medical image, as in the case the feature is intensity, or may require an additional operation of processing or computation, for example, as in the case the feature is gradients. The 3D medical image may be obtained, for example, via provided input, uploaded or downloaded via the internet. In some embodiments the 3D image may be obtained by capturing the medical image via a medical imaging device.

(66) Next, a processor segments the 3D medical image to define one or more ROIs (e.g., portions of spine regions), at 3D medical image segmentation step **504**.

(67) At a ROI feature threshold value determination step **506**, the processor determines at least one ROI features threshold, such as image intensity threshold value. At a background feature threshold value determination step **508**, the processor determines a background features threshold, such as image background intensity threshold value.

(68) The determination of the at least one ROI feature threshold and of the background feature threshold of ROI feature threshold value determination step **506** and background feature threshold value determination step **508** may include receiving input values (e.g., from the user) for the at least one ROI feature threshold or for the background feature threshold or for both. In some implementations, the receiving of the input values from the user comprises generating a GUI element to be displayed to the user, such as ROI TH sliders **302**, **312**, **322** and background TH sliders **304**, **314**, **324** of FIGS. 3A-3C. The GUI element may allow the user to adjust the input values, while the 3D rendering and displaying of the 3D rendering to the user are iteratively performed in correspondence to the user adjustment of the input values. In some embodiments, the one or more ROI feature thresholds and the background feature threshold are predetermined. In some embodiments, a default value for some or for each of the one or more ROI feature thresholds and the background feature threshold is predetermined. A 3D rendering based on these default values may be displayed to the user as a default 3D rendering. The user may then adjust the thresholds values as described above.

(69) At model feature threshold value selection step **510**, the processor selects or determines the threshold to be input to a 3D model generation algorithm based on the determined one or more ROI feature thresholds and the background feature threshold. In some embodiments, the lowest ROI feature threshold may be selected or determined as the input threshold for the 3D model generation. In some embodiments, the lowest of the one or more ROI feature thresholds and the background feature threshold may be selected.

(70) At threshold values checking step **512**, the processor checks if the background feature threshold value is greater than the threshold value selected to be provided to the 3D model generation algorithm (model feature threshold value selection step **510** above).

(71) If the result of the check performed in threshold values checking step **512** is positive, i.e., the background feature threshold value is greater than the threshold value selected in model feature threshold value selection step **510**, then selected feature values in the background (e.g., feature values associated with voxels segmented as background (3D) medical image segmentation step **504** above)) are reduced or suppressed in a step **514**. Background feature values which satisfy the model generation threshold (i.e., the threshold selected to be input to the 3D model generation algorithm in model feature threshold value selection step **510** above) but do not satisfy the background feature threshold are reduced to a value equal to or below the model generation threshold. In some embodiments, such background feature values are reduced to a value equal to the value of the model generation threshold minus a constant. Thus, background elements (e.g., voxels) having feature values which do not satisfy the background feature threshold, are not considered for the model generation, or excluded from being included in the model generation algorithm input.

(72) At a step **516**, a 3D model is generated by a model generation algorithm that accepts a single 3D image feature threshold value selected at model feature threshold value selection step **510**. An example of such a model is the aforementioned marching cubes algorithm.

(73) Finally, the processor displays (or generates an output for display) the generated 3D model to the user, at a 3D model displaying step **518**.

(74) The flowchart of FIG. 5 is presented by way of example and is simplified for clarity of presentation. The process may include alternative steps, such as determining threshold values that are from different medical modalities.

(75) FIGS. **6A-6D** are example GUI displays **600** that include various views of a patient's spine, in accordance with an embodiment of the disclosure. FIGS. **6A-6D** show an illustration of image data displayed during a medical procedure for inserting pedicle screws into a patient's spine. The image data include a display of a 3D model or rendering **601** of an ROI (e.g., a portion of the patient's spine), and virtual images of a tool **602** and a pedicle screw implant **604**. In some embodiments, 3D model **601** may be generated as disclosed hereinabove. The GUI displays **600** in FIGS. **6A-6D** further include various views, represented as images, of the patient spine and pedicle screw implant **604**: X-ray lateral view **606**, X-ray anteroposterior (AP) view **608**, 3D lateral view **610** and 3D AP view **612**, displayed in different combinations.

(76) The displays shown in FIGS. **6A-6D** are not augmented-reality displays and may be displayed, for example, on a workstation. However, these displays may be displayed in a see-through display of a wearable, hands-free and/or head-mounted display system while 3D model **601** is overlaid on the patient anatomy, optionally in-alignment, and as described hereinabove. The different views (**606**, **608**, **610** and/or **612**) may be displayed on a different section of the see-through display, e.g., in a top portion or a side portion of the display and such that they will not interfere with the user's view (e.g., direct see-through view) of the physical ROI through the display.

(77) A user may select which view will be presented in each view display window, such as windows **614** and **616**. The different views may be presented in the GUI display, for example, via a drop-down menu, such as drop-down menu **618**, associated with each view display window. The GUI display **600** shown in FIGS. **6A-6D** includes two view display windows. Other embodiments may include a different number of view display windows. In some embodiments, the user may select which view to display in which view display window (e.g., via a drop-down menu) and switch between the views during the procedure, as desired. Accordingly, the user may generate different combinations of views during the procedure, allowing him or her to receive maximal and proper information for each maneuver, occurrence and/or phase of the procedure.

(78) X-ray views **606** and **608** are X-ray like images or virtual X-ray images derived from the 3D

medical image as viewed from a fixed position, specifically lateral and AP respectively. Lateral and AP X-ray like or virtual X-ray views may facilitate navigation while AP view may allow, for example, symmetric screw insertion planning (e.g., during minimally invasive surgical (MIS) procedures). The virtual X-ray images may be generated by the processor from the 3D medical image by generating digitally reconstructing radiographs. In some embodiments, the processor may utilize the segmentation to digitally reconstruct radiographs from the segmented 3D medical image. The voxel feature values segmented as background (i.e., not ROI) may be set to zero, thus generating X-ray views of the ROI only (e.g., without noise or with reduced noise), as shown, for example, in FIGS. 6A-6D. In some embodiments, a GUI element such as slider **620** may be generated to allow the user to adjust the DRR images visibility threshold (e.g., sum of voxels intensity values). Each change of visibility threshold may cause the processor to generate a new DRR based on the adjusted visibility threshold.

(79) In some embodiments, X-ray views **606** and **608** may display 2D X-ray images of the patient anatomy captured by an X-ray imaging device such as a fluoroscope. In some embodiments, the X-ray images may be registered with the 3D medical image.

(80) 3D views **610** and **612** display the 3D model as viewed from a fixed position, specifically lateral and AP respectively. The 3D view may facilitate visualization and orientation and may simplify entry point location.

(81) Although FIGS. 6A-6D show X-ray views and 3D views from lateral and AP points of view, other points of view or angles of view may be displayed alternatively or additionally. It should be noted that lateral and AP points of view may be specifically advantageous for medical procedures or interventions which the professional may perform via a lateral approach (e.g., by positioning the patient on the side of his body), such as a Lateral Lumbar Interbody Fusion (LLIF) procedure.

(82) In some embodiments, as indicated above, the user may select which views will be displayed in each window of the display, such as windows **614** and **616**. FIGS. 6A-6D show such various combinations which may be advantageous to the user during a medical procedure or intervention (e.g., surgical or non-surgical therapeutic or diagnostic procedure). FIG. 6A shows a combination of X-ray views **606** and **608** from different points of view, specifically lateral and AP. FIG. 6B shows a combination of 3D views **610** and **612** from different points of view, specifically lateral and AP. FIG. 6C shows a combination of lateral views **606** and **610** via different types of view (e.g., imaging modalities or techniques), specifically X-ray and 3D, respectively. FIG. 6D shows a combination of AP views **608** and **612** via different types of view, specifically X-ray and 3D, respectively. It should be noted that 3D view may provide surface-related data of a bone structure, while X-ray view may provide depth-related data of the bone structure. Displaying a specific type of view, e.g., X-ray view or 3D view, from multiple points of view or a specific point of view, e.g., lateral or AP, via different types of view at the same time may be advantageous and may facilitate the procedure or intervention or even better the procedure or intervention outcome by providing the required information and/or more information and better visualization of the ROI to the professional performing the procedure or intervention.

(83) While images of views **606**, **608**, **610** and **612** are static, virtual images of the tool **602** and screw implant **604** may be dynamically augmented on images of views **606**, **608**, **610** and **612** presenting the actual location and navigation of the tool and/or screw implant with respect to the patient anatomy. Such display may be facilitated via a tracking system and as described hereinabove. Since screw implant **604** is attached as a straight extension to tool **602** and since its dimensions may be predefined, there is no need for tracking of screw implant **604** and tracking of only tool **602** may suffice.

(84) In some embodiments, the GUI may include one or more slice views of the 3D medical image. In some embodiments, a sagittal slice view and an axial slice view may be generated and displayed. A virtual image representing a tool and/or a virtual image representing an implant, such as a pedicle screw may be augmented on the slice views, while navigated by a professional during a medical

procedure or intervention. In some embodiments, the displayed slice may be selected and/or generated based on or according to the position of the navigated tool. In some embodiments, the displayed slice may be selected and/or generated such that a predefined axis of the tool is aligned with the slice plane. For example, for an elongated tool, such as a screwdriver, the tool longitudinal axis may be predefined as the tool axis. Thus, for each tracked and/or recorded change of position of the tool a new slice may be generated and displayed.

(85) In some embodiments, for each anatomical view only slices of the specific anatomical view would be generated and displayed in response to the manipulation of the tool. For example, for an axial view, only axial slices would be displayed according to the axial component of the tool current position. In some embodiments, in one or more anatomical views, slices may be generated based on tool movement in two or three anatomical planes. For example, in an axial view, for a current position of the tool having components in the axial and coronal planes, a slice positioned axially and coronally in accordance with the axial and coronal tool position, would be generated and displayed. As another example, in an axial view, for a current position of the tool having components in the axial, coronal and sagittal planes, a corresponding slice would be generated for the axial view. Thus, for example, for a tool positioned in a sagittal plane, the axial slice view and the sagittal slice view may be similar or even identical. Such anatomical slice view which allows slice generation in multiple plains and not just in the predefined view plane provides better visualization and may facilitate navigation.

(86) In some embodiments, the display of the various views may be manipulated by a user. In some embodiments a view may be mirrored, e.g., with respect to an anatomical plane of the patient. For example, a mirroring command button may be included in the GUI. Mirroring of a lateral view may be performed, for example, with respect to a plane of the patient body. In some embodiments, the view may be rotated by a specific angle, e.g., by 180°, with respect to an anatomical axis, for example. Such rotating may be performed by pressing a dedicated command button generated as part of the GUI, for example. Specifically, such rotating of an AP view may provide a view rotated by 180° with respect to the longitudinal anatomical axis (i.e., head to toe). Such manipulation may be advantageous, for example, when the medical image and its derived images are displayed from a side opposite to the desired side (e.g., left side or right side of the patient or upper side or lower side of the patient). In some embodiments, the various views may be panned and zoomed in and out, e.g., via dedicated command buttons generated as part of the GUI.

(87) While examples of the disclosed technique are given for body portion containing spine vertebrae, the principles of the system, method, and/or disclosure may also be applied to other bones and/or body portions than spine, including hip bones, pelvic bones, leg bones, arm bones, ankle bones, foot bones, shoulder bones, cranial bones, oral and maxillofacial bones, sacroiliac joints, etc.

(88) The disclosed technique is presented with relation to image-guided surgery systems or methods, in general, and accordingly, the disclosed technique of visualization of medical images should not be considered limited only to augmented reality systems and/or head-mounted systems. For example, the technique is applicable to the processing of images from different imaging modalities, as described above, for use in diagnostics.

(89) The terms “top,” “bottom,” “first,” “second,” “upper,” “lower,” “height,” “width,” “length,” “end,” “side,” “horizontal,” “vertical,” and similar terms may be used herein; it should be understood that these terms have reference only to the structures shown in the figures and are utilized only to facilitate describing embodiments of the disclosure. Various embodiments of the disclosure have been presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the disclosure. The ranges disclosed herein encompass any and all overlap, sub-ranges, and combinations thereof, as well as individual numerical values within that range. For example, description of a range such as from about 5 to about 30 degrees should be

considered to have specifically disclosed subranges such as from 5 to 10 degrees, from 10 to 20 degrees, from 5 to 25 degrees, from 15 to 30 degrees etc., as well as individual numbers within that range (for example, 5, 10, 15, 20, 25, 12, 15.5 and any whole and partial increments therebetween). Language such as “up to,” “at least,” “greater than,” “less than,” “between,” and the like includes the number recited. Numbers preceded by a term such as “about” or “approximately” include the recited numbers. For example, “approximately 2 mm” includes “2 mm.” The terms “approximately”, “about”, and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result.

(90) In some embodiments, the system comprises various features that are present as single features (as opposed to multiple features). For example, in one embodiment, the system includes a single HMD, a single camera, a single processor, a single display, a single fiducial marker, a single imaging device, etc. Multiple features or components are provided in alternate embodiments.

(91) In some embodiments, the system comprises one or more of the following: means for imaging (e.g., a camera or fluoroscope or MRI machine or CT machine), means for calibration or registration (e.g., adapters, markers, objects), means for fastening (e.g., anchors, adhesives, clamps, pins), etc.

(92) The processors described herein may include one or more central processing units (CPUs) or processors or microprocessors. The processors may be communicatively coupled to one or more memory units, such as random-access memory (RAM) for temporary storage of information, one or more read only memory (ROM) for permanent storage of information, and one or more mass storage devices, such as a hard drive, diskette, solid state drive, or optical media storage device. The processors (or memory units communicatively coupled thereto) may include modules comprising program instructions or algorithm steps configured for execution by the processors to perform any of all of the processes or algorithms discussed herein. The processors may be communicatively coupled to external devices (e.g., display devices, data storage devices, databases, servers, etc. over a network via a network communications interface.

(93) In general, the algorithms or processes described herein can be implemented by logic embodied in hardware or firmware, or by a collection of software instructions, possibly having entry and exit points, written in a programming language, such as, for example, Python, Java, Lua, C, C #, or C++. A software module or product may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language such as, for example, BASIC, Perl, or Python. It will be appreciated that software modules may be callable from other modules or from themselves, and/or may be invoked in response to detected events or interrupts. Software modules configured for execution on computing devices may be provided on a computer readable medium, such as a compact disc, digital video disc, flash drive, or any other tangible medium. Such software code may be stored, partially or fully, on a memory device of the executing computing device, such as the processing system 50, for execution by the computing device. Software instructions may be embedded in firmware, such as an EPROM. It will be further appreciated that hardware modules may be comprised of connected logic units, such as gates and flip-flops, and/or may be comprised of programmable units, such as programmable gate arrays or processors. The modules described herein are preferably implemented as software modules but may be represented in hardware or firmware. Generally, any modules or programs or flowcharts described herein may refer to logical modules that may be combined with other modules or divided into sub-modules despite their physical organization or storage.

(94) The various features and processes described above may be used independently of one another or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure. In addition, certain method or process blocks or steps may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks, steps, or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks, steps, or states may be

performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks, steps, or states may be performed in serial, in parallel, or in some other manner. Blocks, steps, or states may be added to or removed from the disclosed example embodiments. The example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed example embodiments.

(95) Any process descriptions, elements, or blocks in the flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process.

(96) It will be appreciated that the systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features and processes described above may be used independently of one another or may be combined in various ways. The section headings used herein are merely provided to enhance readability and are not intended to limit the scope of the embodiments disclosed in a particular section to the features or elements disclosed in that section.

(97) Certain features that are described in this specification in the context of separate embodiments also may be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment also may be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination. No single feature or group of features is necessary or indispensable to each and every embodiment.

(98) Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

(99) The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. In addition, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise.

Claims

1. A computer-implemented method for improving display of 3D models in connection with image-guided surgery, the method comprising: obtaining a three-dimensional (3D) computed tomography image of a region of a spine of a patient, the 3D computed tomography image having intensity values; segmenting the 3D computed tomography image to define multiple regions of interest (ROIs) of the spine and a background region, wherein the background region comprises a portion of the 3D computed tomography image which is not an ROI; determining, for each of the multiple ROIs of the spine, a ROI intensity threshold value that can be used as an input to control what portions of the ROI are included in a 3D rendering of the 3D computed tomography image;

determining a background intensity threshold value that can be used as an input to control what portions of the background region are included in the 3D rendering of the 3D computed tomography image; generating the 3D rendering of the 3D computed tomography image, wherein the generation of the 3D rendering comprises: in each of the multiple ROIs of the spine of the 3D computed tomography image, rendering based on intensity values of the ROI that satisfy a lowest threshold value of the determined ROI intensity threshold value and the determined background intensity threshold value; and in the background region of the 3D computed tomography image, rendering based on intensity values of the background region that satisfy the determined background intensity threshold value; and causing the 3D rendering to be output for display to a user, wherein the intensity threshold values are voxel values.

2. The computer-implemented method of claim 1, further comprising displaying the 3D rendering in alignment by performing registration of the 3D rendering with the region of the spine.

3. The computer-implemented method of claim 1, wherein the 3D rendering is a 3D model.

4. The computer-implemented method of claim 1, wherein the generating the 3D rendering comprises changing at least some of the intensity values of the 3D computed tomography image into a value which does not satisfy the lowest threshold value.

5. The computer-implemented method of claim 1, wherein the background region of the 3D computed tomography image comprises soft tissue.

6. The computer-implemented method of claim 1, further comprising: repeatedly adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user; and repeatedly generating the 3D rendering of the 3D computed tomography image based on the adjusted values.

7. The computer-implemented method of claim 1, wherein the multiple ROIs of the spine comprise various regions of the spine.

8. The computer-implemented method of claim 1, wherein the multiple ROIs of the spine comprise individual vertebrae of the spine.

9. The computer-implemented method of claim 1, further comprising: adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user; regenerating the 3D rendering of the 3D computed tomography image based on the adjusted values; and generating a 3D model based on the adjusted values.

10. A computer-implemented method for improving display of 3D models in connection with image-guided surgery, the method comprising: obtaining a three-dimensional (3D) image of a region of a spine of a patient, the 3D image having intensity values, segmenting the 3D image to define multiple regions of interest (ROIs) of the spine and a background region, wherein the background region comprises a portion of the 3D image which is not an ROI; determining, for each of the multiple ROIs of the spine, a ROI intensity threshold value that can be used as an input to control what portions of the ROI are included in a 3D rendering of the 3D image; determining a background intensity threshold value that can be used as an input to control what portions of the background region are included in the 3D rendering of the 3D image; generating the 3D rendering of the 3D image, wherein the generation of the 3D rendering comprises: in each of the multiple ROIs of the spine of the 3D image, rendering based on intensity values of ROI that satisfy a lowest threshold value of the determined ROI intensity threshold value and the determined background intensity threshold value; and in the background region of the 3D image, rendering based on intensity values of the background region that satisfy the determined background intensity threshold value; and causing the 3D rendering to be output for display to a user.

11. The computer-implemented method of claim 10, wherein the intensity values are voxel values of the 3D image.

12. The computer-implemented method of claim 10, wherein the 3D image is a computed tomography image or a magnetic resonance image.

13. The computer-implemented method of claim 10, further comprising displaying the 3D

- rendering in alignment by performing registration of the 3D rendering with the region of the spine.
14. The computer-implemented method of claim 10, wherein the 3D rendering is a 3D model.
 15. The computer-implemented method of claim 10, wherein the generating the 3D rendering comprises changing at least some of the intensity values of the 3D image into a value which does not satisfy the lowest threshold value.
 16. The computer-implemented method of claim 10, wherein the background region of the 3D image comprises soft tissue.
 17. The computer-implemented method of claim 10, further comprising: repeatedly adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user; and repeatedly generating the 3D rendering of the 3D image based on the adjusted values.
 18. The computer-implemented method of claim 10, further comprising: adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user; regenerating the 3D rendering of the 3D image based on the adjusted values; and generating a 3D model based on the adjusted values.
 19. The computer-implemented method of claim 10, wherein the multiple ROIs of the spine comprise various regions of the spine.
 20. The computer-implemented method of claim 10, wherein the multiple ROIs of the spine comprise individual vertebrae of the spine.
-